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## Economics



# **INTRODUCTION TO ECONOMICS**

# **CHAPTER ONE**

## **FUNDAMENTALS OF ECONOMICS**

# 1.1. DEFINITION AND NATURE OF ECONOMICS

❖ The word economics comes from the ancient Greek word ***oikonomia***.

*oikos* => *house*

*nomos* => *rule or custom*

*So oikonomia means rule of house(household) or management of household administration.*

❖ Two fundamental facts together provide a foundation for Economics:

1. Society's material want are unlimited.
2. Economic resources are limited in supply or scarce

## Cont...

- ❖ *By society's material wants* we refer to the desire of consumers, business (firms), and government to get those things that help them realize their respective goals.
- ❖ The goal of the consumer is to get *maximum satisfaction*, the goal of the business is to produce goods and services for *profit* and the goal of the government is to satisfy the *collective wants of its citizens*.

## Cont...

- ❖ Human wants are not only numerous but also expand and diversify through time. Therefore, human wants are unlimited.
- ❖ **Resource** is anything natural or man made that can be used in production of goods and services.
- ❖ Thus, **economic resources** are the means to produce goods and services. Examples are various types of labor, oil deposits, minerals, building, communication facilities etc.
- ❖ All these resources are scarce or limited in supply.

## Cont...

- ❖ These two contradictory facts lay the foundation for the field of Economics
- ❖ ***Economics*** is thus a science which studies the allocation of scarce resources in production, consumption and distribution of goods and services to attain the maximum fulfillment of society's material wants.
- ❖ Economics is concerned with “doing the best with what we have”.

# Cont...

❖ From the above definition we understand the following points.

1. Economics is Science
2. The society faces a problem of scarcity
3. There is a problem of choice



# The scope of Economics

- ❖ From the point of view of elements of analysis, economics has two major branches: **microeconomics and macroeconomics**

## A. Microeconomics

- Deals with economic behaviors of individual economic units such as consumers, producers, business firms and other economic decision making units
- Deals with how the market of individual commodities function

# Cont...

- ❖ It is concerned also with interaction among the economic units

## **B. Macroeconomics**

- ❖ Is branch of economic analysis concerned with the economy as a whole and sub aggregates of the economy.
- ❖ Macroeconomics deals with aggregate units of national economy such as national output or Gross National Product (GDP), general price level, inflation and national employment.

## Cont...

- Like macroeconomics ,microeconomics also uses aggregates. But in microeconomics we aggregate over the homogeneous product but in macroeconomics the aggregation is at the economy level

# Basic Classifications of Economics

## Macroeconomics

The big picture. Study of the operation of the economy as a whole. It looks at aggregate data.



Focuses more at the policy and regulatory levels.

## Microeconomics

“A small-scale study”. Focuses on individual entities of the economy, such as households and firms.

focuses at the firm level



- Both areas offer valuable outlook on the economy.

Table 1. Comparison between Microeconomics and Macroeconomics

| Microeconomics                                | Macroeconomics               |
|-----------------------------------------------|------------------------------|
| ✓ Households' and/or firms' economic behavior | ✓ National economy           |
| ✓ Price of individual items or firms          | ✓ General price level        |
| ✓ Price of a consumption good                 | ✓ Consumer price index (CPI) |
| ✓ An increase in price of a commodity         | ✓ Inflation                  |
| Output of a factory or a firm                 | ✓ National output or GDP     |

# Cont...

| Microeconomics                             | Macroeconomics                         |
|--------------------------------------------|----------------------------------------|
| ✓ Output of a factory or a firm            | ✓ National output or GDP               |
| ✓ Wage rate in an enterprise               | ✓ National wage rate                   |
| ✓ Investment of individual firm            | ✓ National Investment                  |
| ✓ Consumption of households or individuals | ✓ National Consumption                 |
| ✓ Firm's expense or                        | ✓ Government expenditure <sup>13</sup> |

# 1.2. Scarcity and Choice

**Scarcity** –refers to all economic resources, that a society needs to produce goods and resources, are finite or limited in supply

- ✓ It refers to the imbalance between our wants and a means to satisfy these wants.
- ✓ Resources can be classified as free and economic resources
- ✓ A resource is said to be free if the amount available to society is greater than the amount people desire at zero price.

## Cont...

- ✓ A resource is said to be scarce or economic resource if the amount available to society is less than the amount people desire at zero price
- ✓ Economic resource can also be divided into four categories
  - i. Labor
  - ii. Land
  - iii. Capital
  - iv. Entrepreneurship



# Cont...

## i. Labor

- ✓ refers to the physical as well as mental effort of human beings in production and distribution of goods
- ✓ This includes both the skilled and unskilled labor
- ✓ The reward for labor is called *wage*

# Cont...

## ii. Land

- ✓ refers to the natural resources or all the free gifts of nature
- ✓ It doesn't include those resources that are transformed, altered or improved by human beings
- ✓ The reward/payment for land is called ***rent***

# Cont...

## iii. Capital

- ✓ refers to all manufactured inputs that can be used to produce other goods and services
- ✓ it refers to physical capital only. It doesn't take in to account financial capital
- ✓ The reward for capital is called *interest*

# Cont...

## iv. Entrepreneurship

- ✓ refers to a special type of human talent that helps to organize, manage factors of production to produce goods and services
- ✓ Takes risk of making losses
- ✓ The reward for an entrepreneur is *profit*

## Cont...

- The same logic(definition of scarce or free resource ) is applicable for goods and services
- A good is said to be scarce or economic good if the amount available to society is less than the amount people desire at zero price
- A good is said to be free good if the amount available to society is greater than the amount people desire at zero price

# Cont...

NB:scarcity doesn't mean shortage

| Shortage             | Scarcity              |
|----------------------|-----------------------|
| Specific problem     | Universal problem     |
| Short term phenomena | Everlasting phenomena |

# Cont...

- ✓ Scarcity implies choice
- ✓ Choice in turn implies cost(sacrificiation)
- ✓ The cost is the forgone opportunity
- **Opportunity cost/opportunity lost/:** is the value of the next best alternative that must be given up in order to obtain one more unit of the first product

# Production Possibility Frontier(PPF)

- ❖ PPF is a graph that shows the various combinations of output that the economy *can possibly* produce given the available factors of production and the available production technology

Assumptions to draw the PPC/PPF

- i. Efficiency
- ii. Fixed resource
- iii. Two products
- iv. Fixed technology

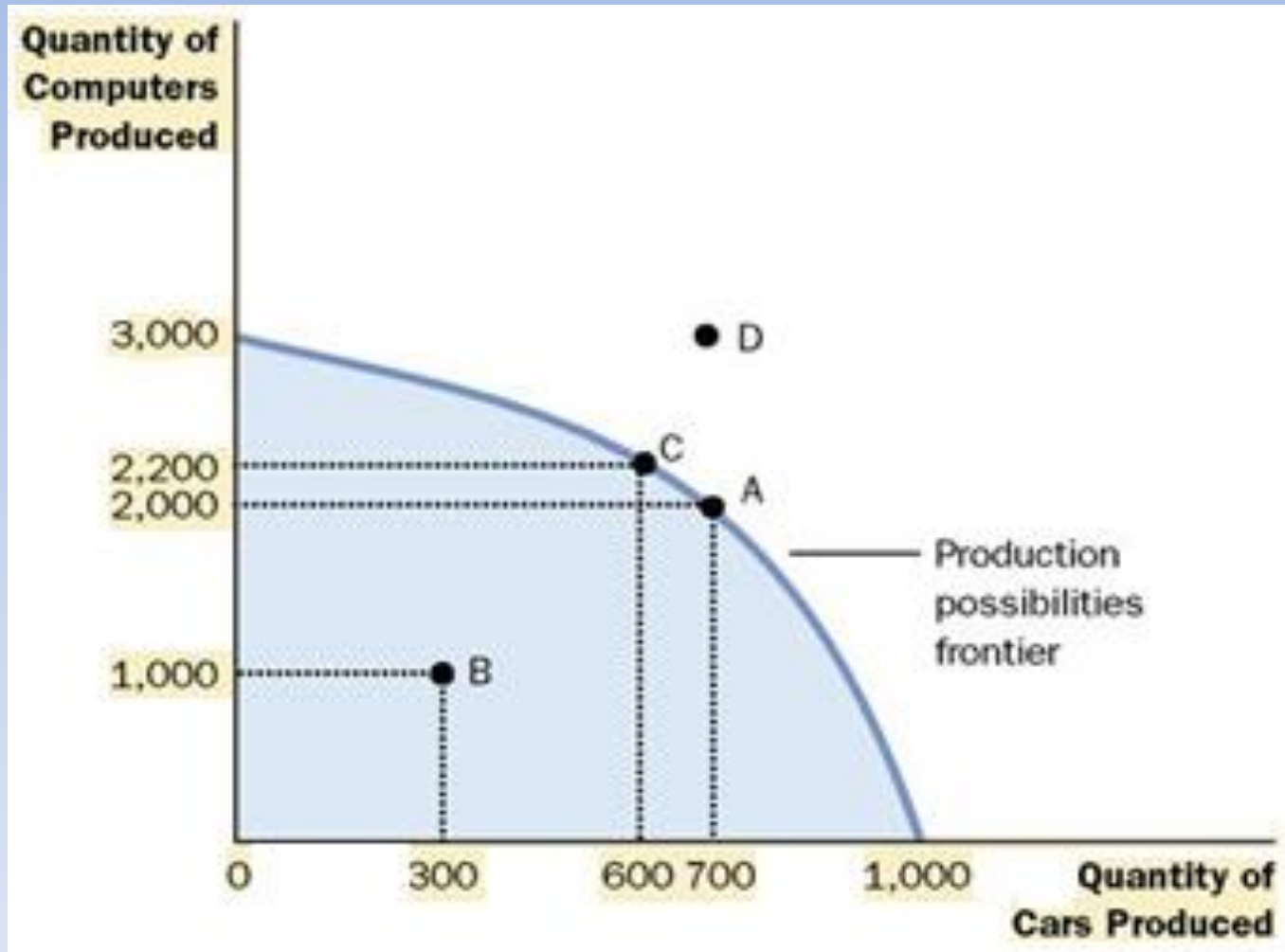


## Cont...

- ❖ Although real economies produce thousands of goods and services, let's imagine an economy that produces only two goods—cars and computers
- ❖ Together the car industry and the computer industry use all of the economy's factors of production

# Cont...

Figure 1



## Cont...

- ❖ Figure 1 is an example of a production possibilities frontier. In this economy, if all resources were used in the car industry, the economy would produce 1,000 cars and no computers
- ❖ If all resources were used in the computer industry, the economy would produce 3,000 computers and no cars

## Cont...

- ❖ The two end points of the production possibilities frontier represent these extreme possibilities
- ❖ If the economy were to divide its resources between the two industries, it could produce 700 cars and 2,000 computers, shown in the figure by point **A**
- ❖ Points on PPF are attainable and efficient

## Cont...

- ❖ By contrast, the outcome at point **D** is not possible because resources are scarce: The economy does not have enough of the factors of production to support that level of output
- ❖ In other words, the economy can produce at any point on or inside the production possibilities frontier(*attainable but inefficient*)
- ❖ the economy cannot produce at points outside the frontier(*Unattainable*)

## Cont...

- ❖ The production possibilities frontier shows one tradeoff that society faces.
- ❖ Once we have reached the efficient points on the frontier, the only way of getting more of one good is to get less of the other. When the economy moves from point A to point C
- ❖ For instance, society produces more computers but at the expense of producing fewer cars

# Cont...

- ❖ Another Principles of Economics that is shown through the production possibility frontier is **opportunity cost**
- ❖ The opportunity cost of a commodity means the amount of a next best alternative that must be sacrificed in order to obtain one more unit of the commodity
- ❖ The production possibilities frontier shows the opportunity cost of one good as measured in terms of the other good
- ❖ When society reallocates some of the factors of production from the car industry to the computer industry, moving the economy from point A to point C, it gives up 100 cars to get 200 additional computers. In other words, when the economy is at point A, the opportunity cost of 200 computers is 100 cars

# Cont...

Opportunity =  $\frac{\text{The amount of the good scarified}}{\text{The amount of the good gained}}$

- ***Law of Increasing Opportunity Cost*** – For each additional unit of a good produced the opportunity cost increases. Why?
- Because the most efficient use of the resource in production of a good is used first.



# Cont...

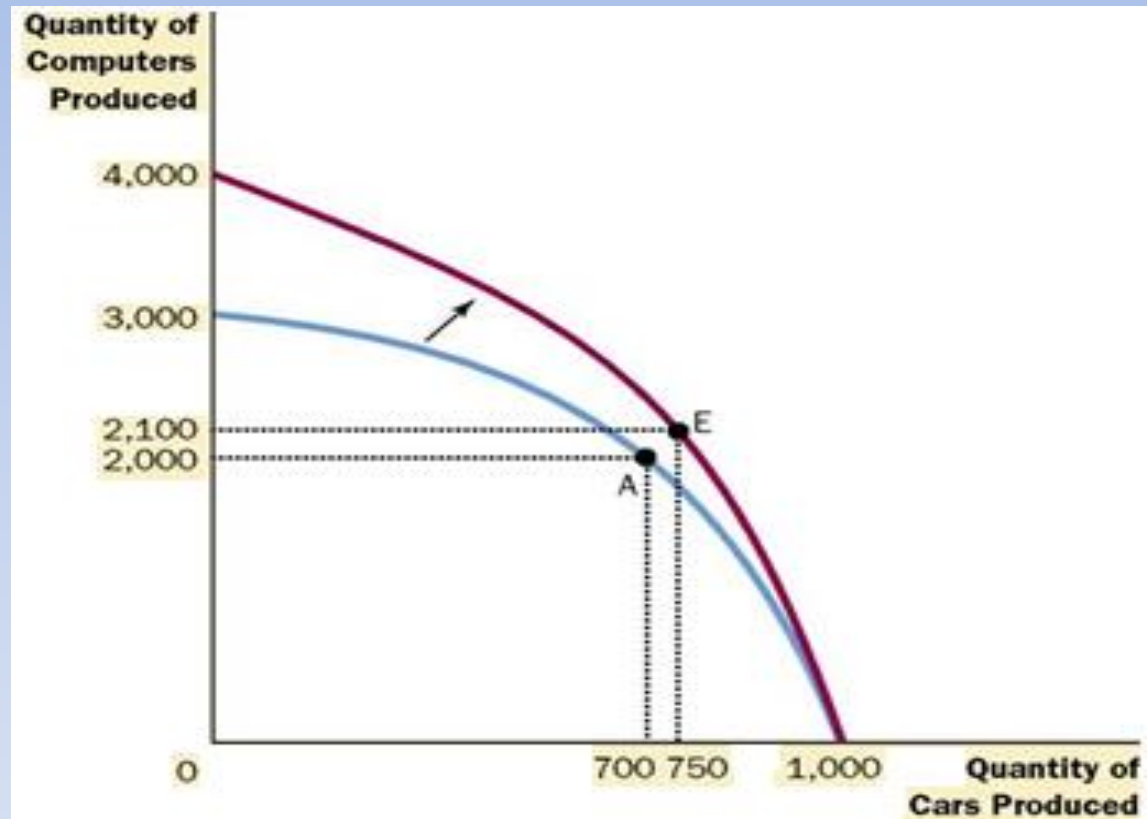
- *Is there any mechanisms to produce outside the production possibility frontier? Or contraction ?*
- Economic Growth (Causes):
  1. Capital accumulation (human, capital goods)
  2. Technological progress (change, ideas)
- Contraction of PPF (Causes):
  1. Droughts
  2. Floods
  3. Earthquakes

## Cont...

❖ For example, if a technological advance in the computer industry raises the number of computers that a worker can produce per week, the economy can make more computers for any given number of cars. These can be presented graphically as follows;

# Cont...

Figure 2



## Cont...

- ❖ As a result, the production possibilities frontier shifts outward, as in Figure 2.
- ❖ Because of this economic growth, society might move production from point A to point E, enjoying more computers and more cars.

# Cont...

- ❖ A change in technology can be symmetrical or asymmetrical
- ✓ If the change is in one sector => Asymmetrical
- ✓ If the change is in both sectors => symmetrical

# 1.3.The Basic Economic Questions and Alternative Economic systems

There are three basic economic problems that any economic unit needs to answer:

- a. *What to produce*:** Types and amounts of commodities to be produced
- b. *How to produce*:** the answer to this question may help determine what production method or technique to use and what input to use. For instance the decision may be about identifying the best combinations of inputs or raw materials

# Cont...

***C. For whom to produce:*** This question helps us to identify potential customers

# Alternative Economic systems

## *Economic system*

- ✓ Is a set of organizational and institutional arrangements
- ✓ Established to answer the basic economic questions
- ✓ There are three types of economic systems
  - a. Market economy
  - b. Command economy
  - c. Mixed economy



# A. Market Economy

- ❖ The private ownership of resources and the use of markets and prices to coordinate and direct economic activity characterize the market system, or capitalism.
- ❖ In this system each participant acts in his or her own self-interest; each individual or business seeks to maximize its satisfaction or profit through its own decisions regarding consumption or production

# Cont...

- ❖ The system allows for the private ownership of capital, communicates through prices, and coordinates economic activity through markets
- ❖ Goods and services are produced and resources are supplied by whoever is willing and able to do so
- ❖ The result is competition among independently acting buyers and sellers of each product and resource

# Summary of Market economy

- **Producers** and **consumer** determine WHAT, HOW, and FOR WHOM to produce.
- ***Advantages =***
  1. *ability to adjust to change;*
  2. *the high degree of individual freedom (start a business, the ability to work nights, part time job, two jobs)*
  3. *the small degree of government involvement;*
  4. *the ability to have a voice in the economy;*
  5. *the variety of goods and services created;*
  6. *the high degree of consumer satisfaction*
  7. *high standard of living*

# Cont...

## *Disadvantages =*

1. inability of the market to meet every person's basic needs \*(market failure)
2. inadequate job in providing highly valued services like justice, education, and health care
3. high level of personal uncertainty and the prospect of economic failure.....failure to stabilize the economy

## B. Command Economic System

- ❖ The alternative to the market system is the command system, also known as socialism or communism.
- ❖ In this system, the government owns most property resources and economic decision making occurs through a central economic plan.
- ❖ A central planning board appointed by the government makes nearly all the major decisions concerning the use of resources, the composition and distribution of output, and the organization of production

## Cont...

- ❖ The government owns most of the business firms, which produce according to government directives.
- ❖ A central planning board determines production goals for each enterprise and specifies the amount of resources to be allocated to each enterprise so that it can reach its production goals

## Cont...

- ❖ The division of output between capital and consumer goods is centrally decided, and capital goods are allocated among industries on the basis of the central planning board's long-term priorities.

# Summary of command Economy

- **A central authority** determines WHAT, HOW and FOR WHOM to produce
- Examples: North Korea, Cuba, People's Republic of China (and the former Soviet Union)

## ***Advantages***

- Fair distribution of income
- Absence of business fluctuation
- Absence of private monopolistic practices

## **Disadvantages =**

1. consumer needs may not be met;
2. hard work is not rewarded;
3. necessary decision-making bureaucracy delays decision;
4. little flexibility to deal with day-to-day problems;
5. individual initiative goes unrewarded (i.e. entrepreneurs need not apply)



## C. Mixed Economic Systems

- Pure capitalism and command economy are the two extreme types of economic systems.
- The mixed economic system takes the strong elements of the two economic systems

## **1.4. Decision making unit and circular flow of economic activities**

- ❖ There are three decision-making units in closed economy.
  - i. Households-is an economic unit which provides an economy with resources and uses the money paid to it to buy goods and services to satisfy its material wants.
  - ii. Firm- a firm is production unit that uses economic resources to produce g + s and sell them to hhs, other firms and gov't.

# Cont'd

Firms makes to decisions:

1. buying of economic resources
  2. Selling of their products
  - iii. Government
- ❖ These economic agents interact in two markets
- i. Resource/input markets
  - ii. Products/output markets

# Cont...

- ❖ We can use two models to understand the economic interaction among economic agents
  - a. Two sector circular flow model
  - b. Three sector circular flow model

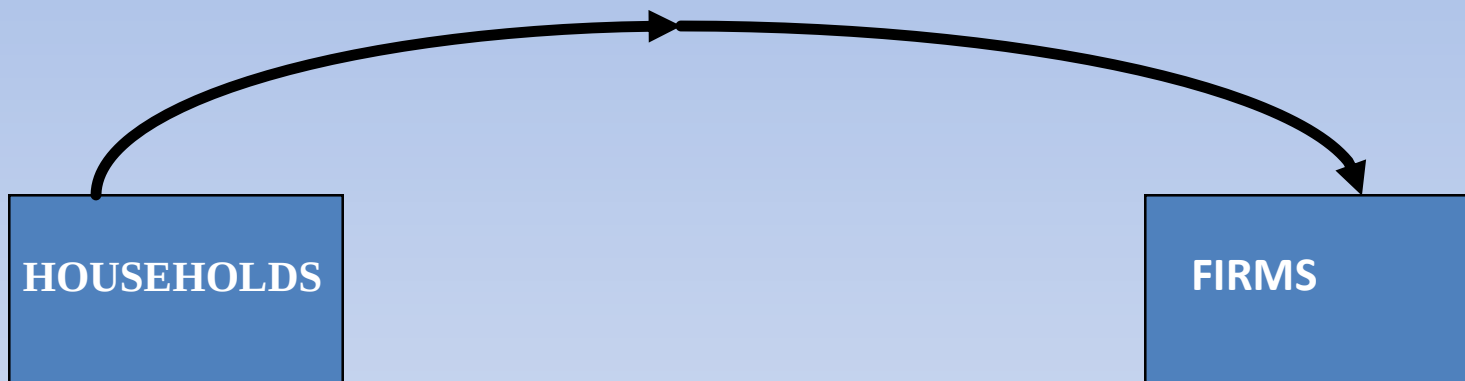
# CIRCULAR FLOW OF ECONOMIC ACTIVITY(TWO SECTOR MODEL)

HOUSEHOLDS

FIRMS

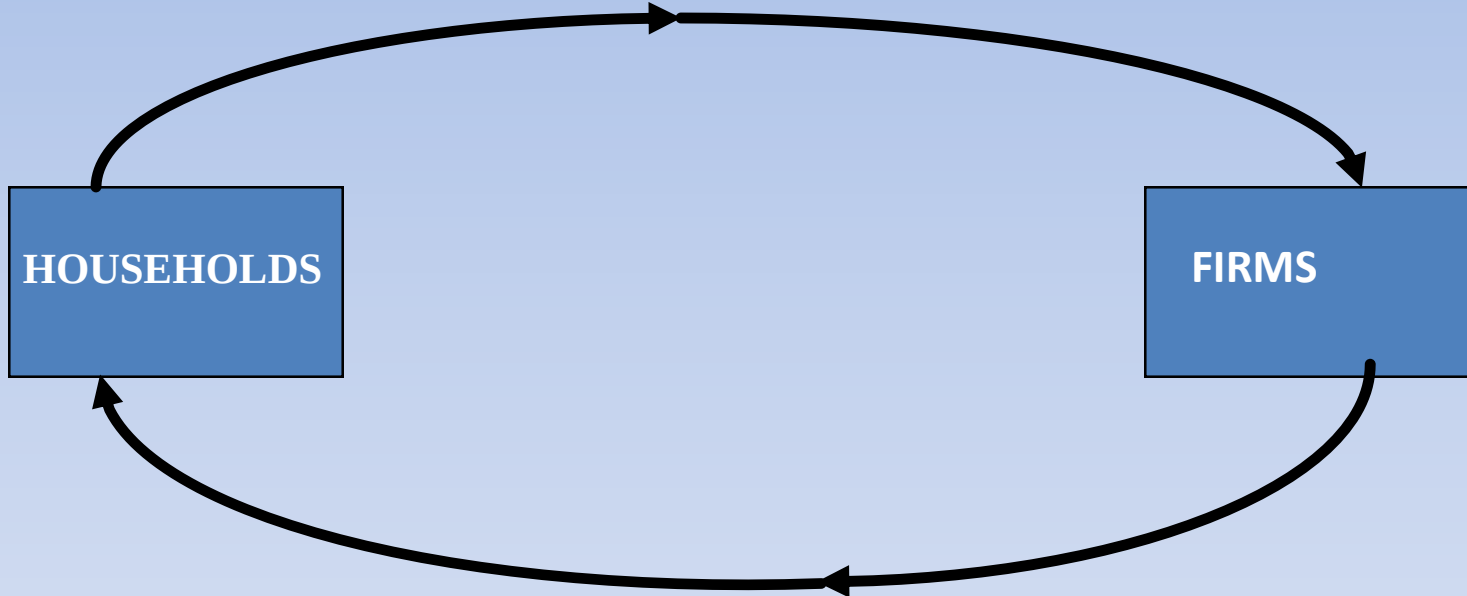
# CIRCULAR FLOW OF ECONOMIC ACTIVITY

PAYMENTS FOR GOODS AND SERVICES



# CIRCULAR FLOW OF ECONOMIC ACTIVITY

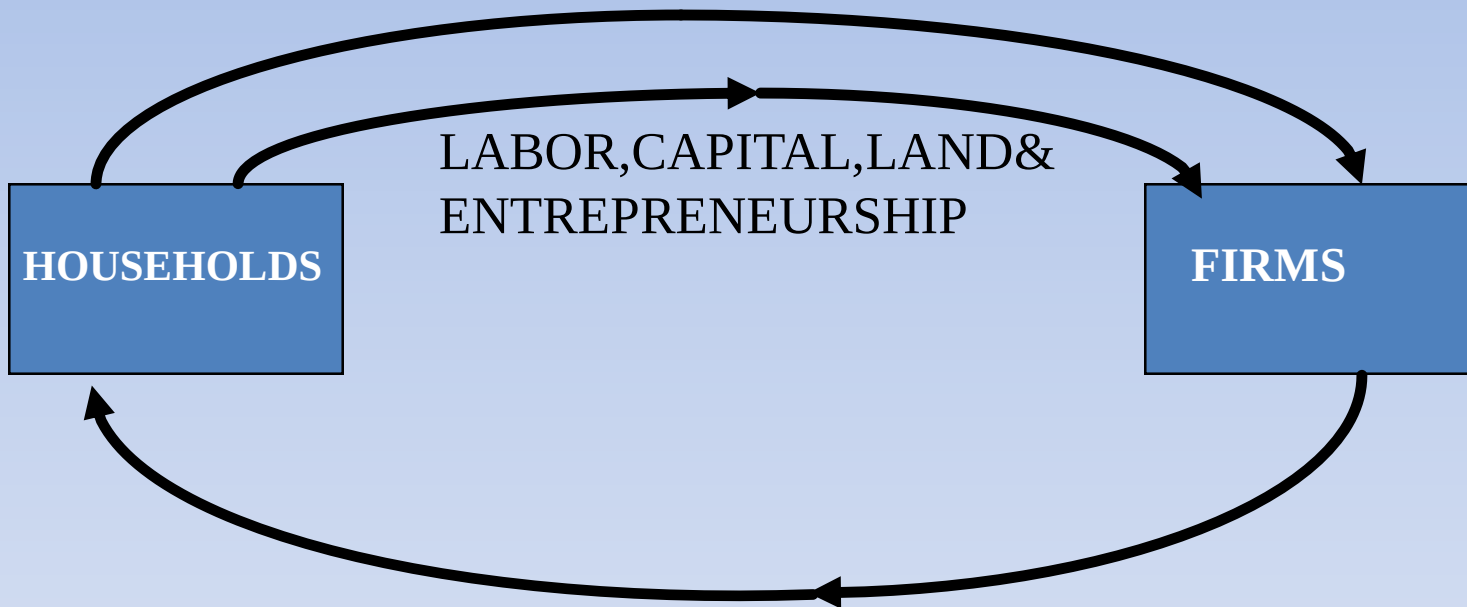
PAYMENTS FOR GOODS AND SERVICES:  
FIRMS' **REVENUE**



WAGES, SALARIES, PROFITS, & RENT:  
FIRMS' **COSTS OR FACTOR PAYMENTS**

# CIRCULAR FLOW OF ECONOMIC ACTIVITY

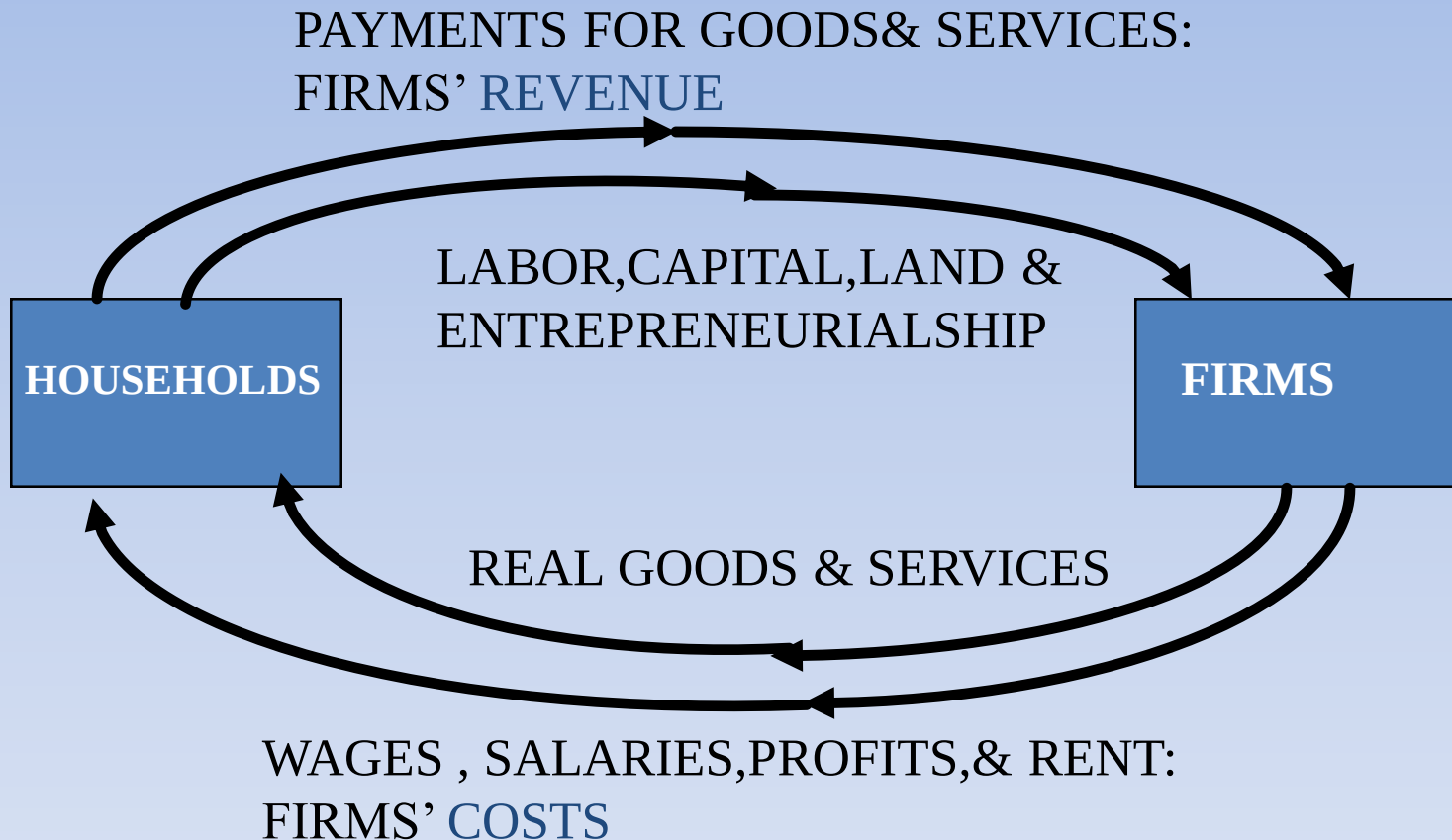
PAYMENTS FOR GOODS AND SERVICES:  
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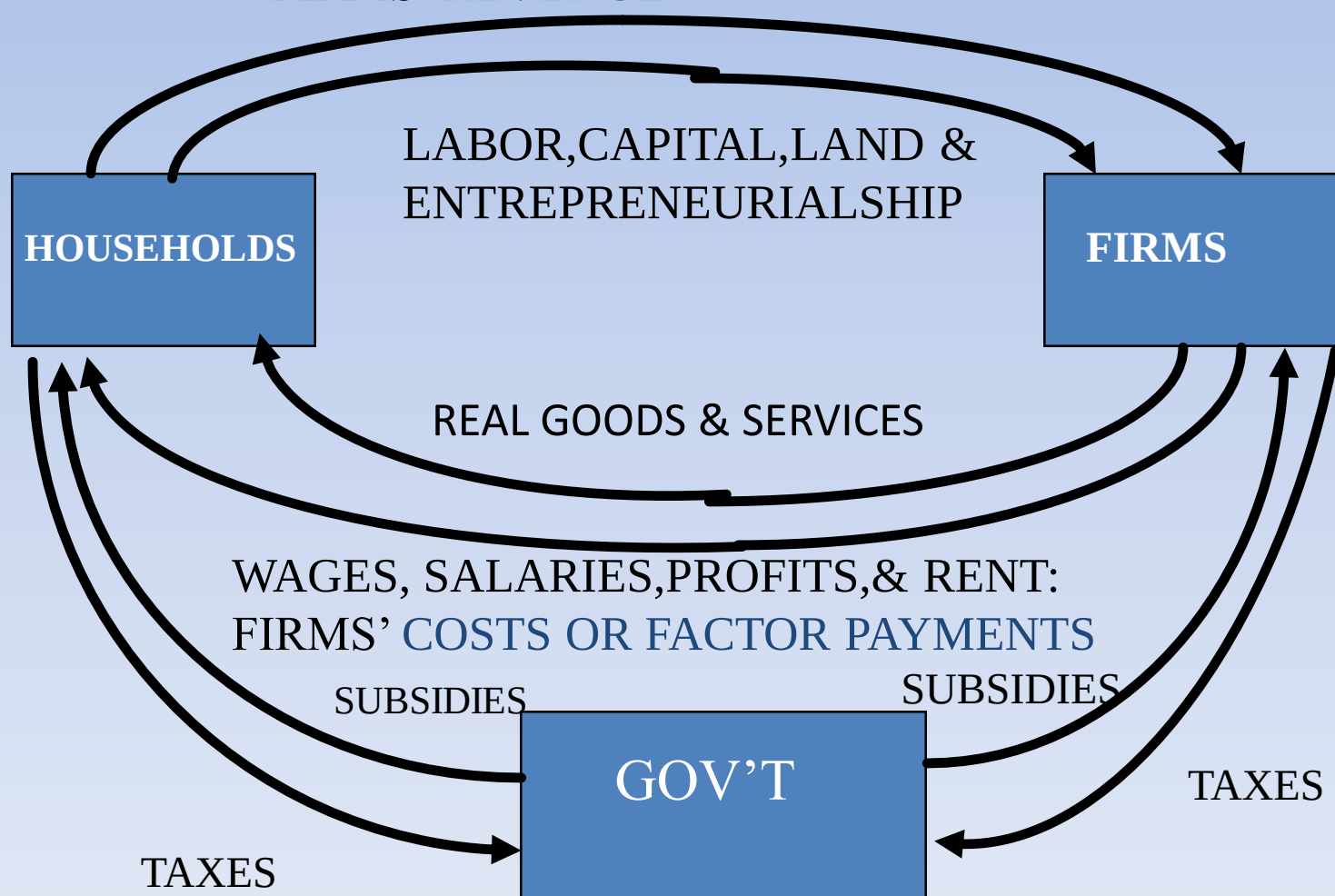


# CIRCULAR FLOW OF ECONOMIC ACTIVITY

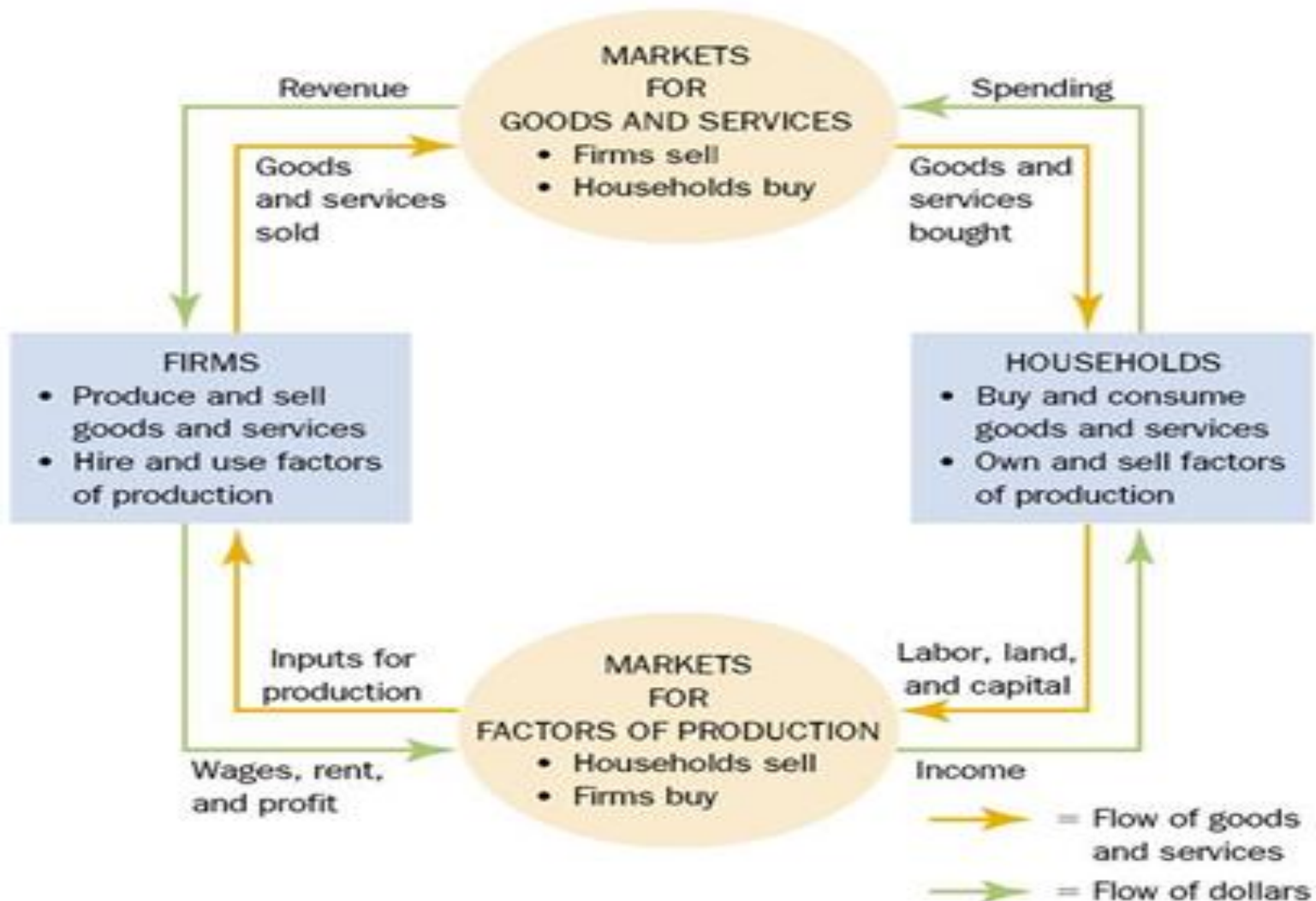


# CIRCULAR FLOW OF ECONOMIC ACTIVITY(THREE SECTOR MODEL)

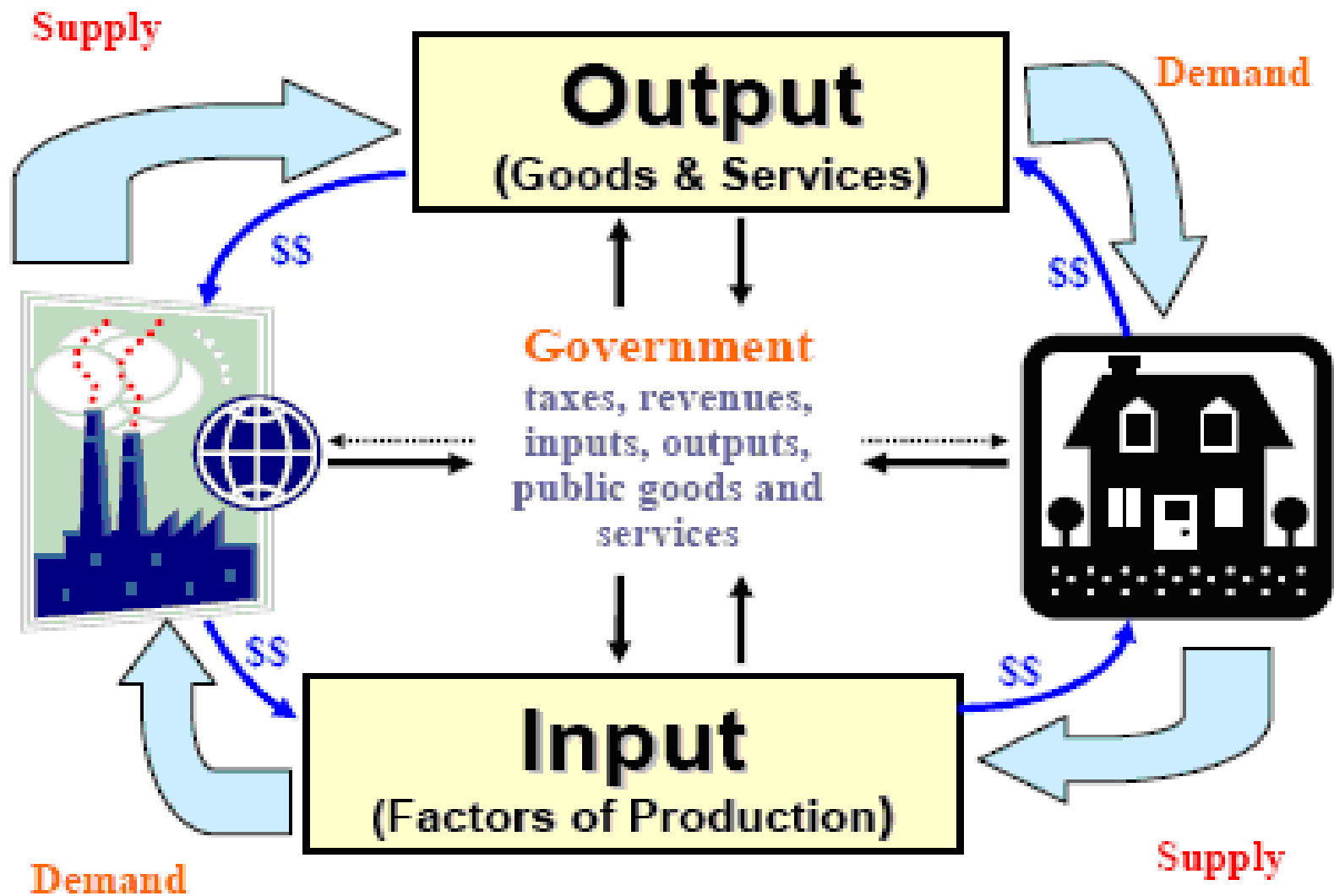
PAYMENTS FOR GOODS& SERVICES:  
FIRMS' **REVENUE**



# Two sector circular flow model



# Three sector circular flow model



**THE END**

# CHAPTER TWO

## THEORY OF DEMAND & SUPPLY AND ELASTICITY

# Demand

- + Quantity Demanded – the amount(number) of a good that a consumer is willing and able to purchase at the given price in a given period of time.
- + Demand –consumers willingness and ability to purchase goods and services at various prices in a specific period of time., holding all else constant.

# Cont...

- ✚ The relationship between price and quantity demanded can be seen using demand schedule, demand curve, demand function.
- ✚ Demand Schedule – a table that shows the relationship between quantity demanded and price ,*ceteris paribus*
- ✚ Demand Curve – a graph that shows the relationship between quantity demanded and price ,*ceteris paribus*
- ✚ Demand Function – a mathematical expression that shows the relationship between quantity demanded and price, *ceteris paribus*
- ✚ Law of Demand – All else equal, as price falls the quantity demanded rises and vice versa.



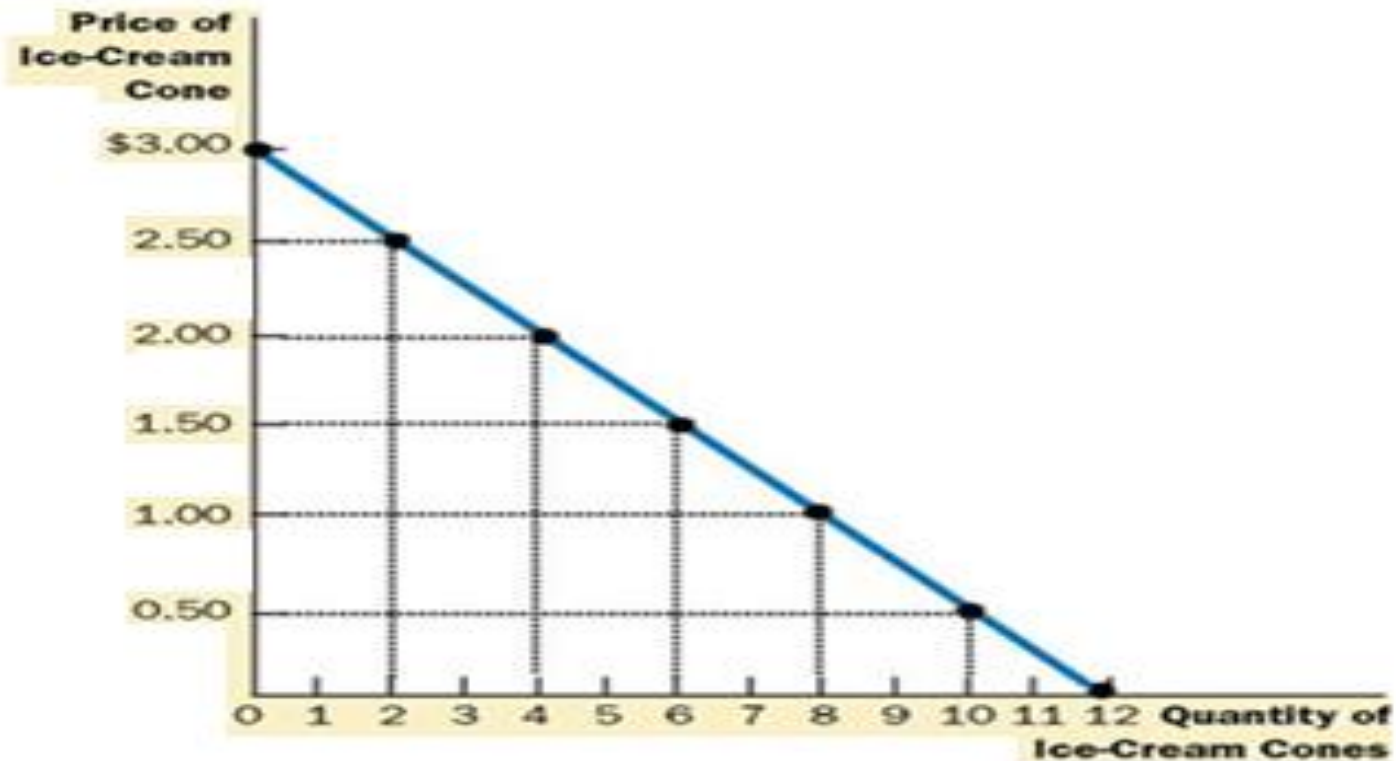
# Cont...

PRICE OF ICE-CREAM CONE

\$0.00  
0.50  
1.00  
1.50  
2.00  
2.50  
3.00

QUANTITY OF CONES DEMANDED

12  
10  
8  
6  
4  
2  
0



# Market Demand Versus Individual Demand Curve

- ➡ Individual Demand Curve – is a curve that represents the price quantity combination of a particular good for a single buyer
- ➡ Market Demand Curve – is a curve that represents the price quantity combination of a particular good for all buyers

# Cont...

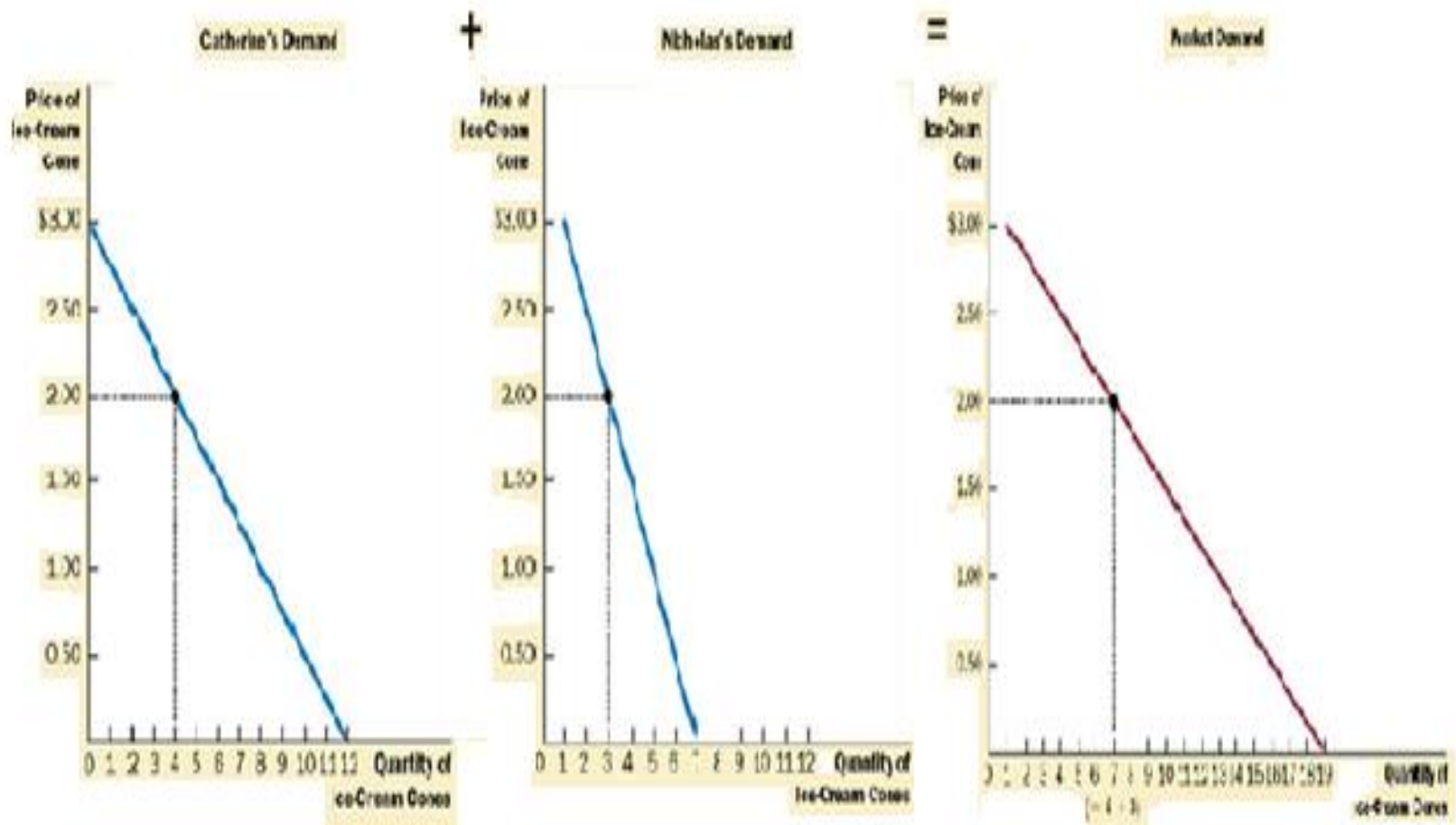
Individual and market demand schedule

| <u>PRICE OF ICE-CREAM CONE</u> | <u>CATHERINE</u> |   | <u>NICHOLAS</u> |   | <u>MARKET</u> |
|--------------------------------|------------------|---|-----------------|---|---------------|
| \$0.00                         | 12               | + | 7               | = | 19            |
| 0.50                           | 10               |   | 6               |   | 16            |
| 1.00                           | 8                |   | 5               |   | 13            |
| 1.50                           | 6                |   | 4               |   | 10            |
| 2.00                           | 4                |   | 3               |   | 7             |
| 2.50                           | 2                |   | 2               |   | 4             |
| 3.00                           | 0                |   | 1               |   | 1             |

# Cont...

Individual and market demand curve

**Exercise:** If there are 1000 identical buyers in the market, each with a demand function of  $Q_{dx}=8-P_x$ , then what is the market demand function?



# Determinants of Demand

1. Price of product itself
2. Income
3. Price of Related Goods
  - a. Substitute Goods
  - b. Complementary Goods
4. Tastes and Preferences
5. Expectations
6. Number of buyers

# Changes in Quantity Demand vs. Changes in Demand

- Changes in Quantity Demand –is represented by a movement along a given demand curve  
=> This is caused by the change in the price of the good
- Changes in Demand –is represented by a shift of the demand curve  
=> This is caused by the change in the determinants of demand other than price of the product

# Cont...

## 1. *Income*

- Normal Good – the higher your income the more you consume

Examples: *Sports Tickets, Cars, and Luxury Goods*

↑Income

=> consume more at each P

=> ↑D

- Inferior Good – as income rises you consume less

Examples: *Shiro wot, Cabbage*

↑Income

=> consume other products

=> ↓ D

# Cont...

## ***2. Price of Related Goods***

➡ Substitute Goods – two goods in which a consumer will consume one good or the other

Examples: *Pepsi or Coke, Rent Movie or Go to Theater*

↑Price of Pepsi

=> drink less Pepsi

=> purchase more Coke

=> ↑D



# Cont...

- Complementary Goods – two goods consumed together

Examples: *gasoil and car, DVD Player and Movie, sugar and tea*

↑ Price of sugar

=> consume less tea

=> consume less sugar and tea

=> ↓ D

# Cont...

## ***3. Tastes / Preference***

New tastes for the product

=> Consume more

=>  $\uparrow D$

## ***4. Expectations***

$\uparrow$ Price tomorrow

=> buy today instead of tomorrow

=>  $\uparrow D$

## ***5. Number of buyers***

$\uparrow$ number of buyers

=> More of the good is consumed

=>  $\uparrow D$

# Supply

- + Quantity Supplied – the amount(number) of a good that a producer is willing and able to offer at each price level in a given period of time.
- + Supply –producers willingness and ability to provide goods and services at different prices in a specific period of time., holding all else constant.

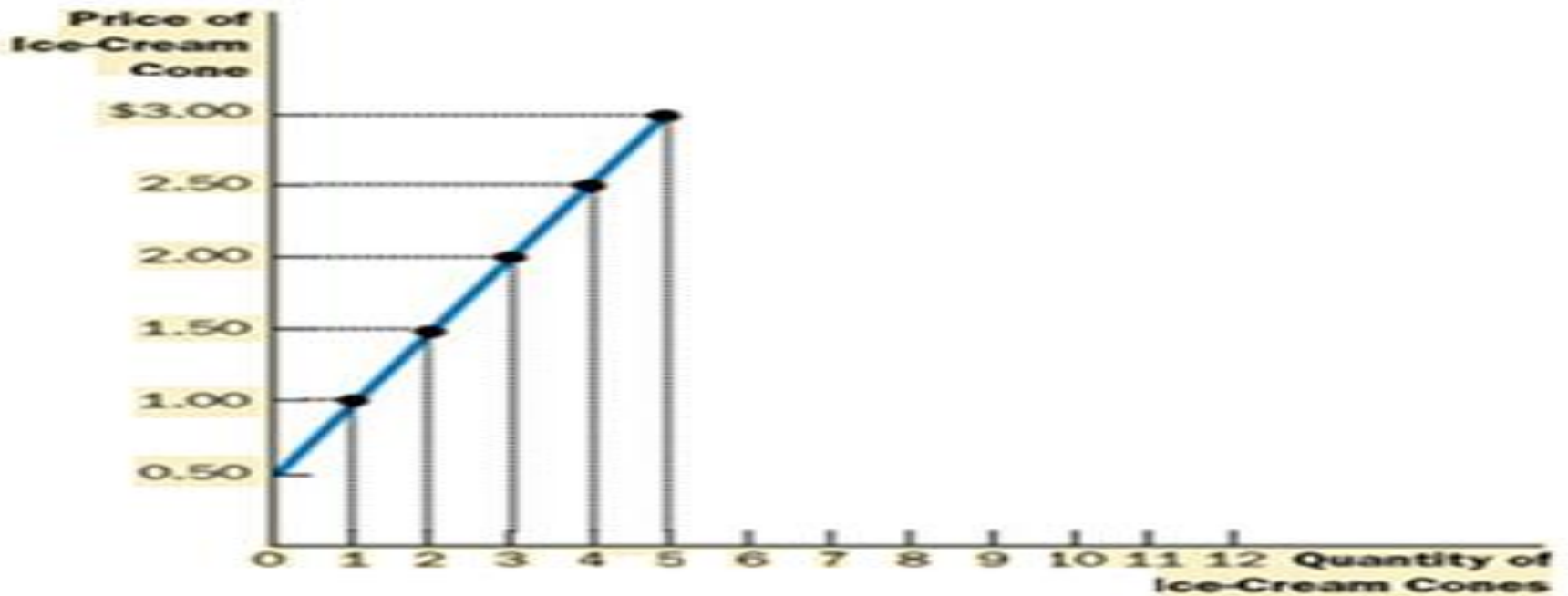
# Cont...

- ✚ The relationship between price and quantity supplied can be seen using supply schedule, supply curve, supply function.
- ✚ Supply Schedule – a table that shows the relationship between quantity supplied and price ,ceteris paribus
- ✚ Supply Curve – a graph that shows the relationship between quantity supplied and price ,ceteris paribus
- ➡ Supply Function – a mathematical expression that shows the relationship between quantity supplied and price,ceteris paribus
- ➡ Law of Supply – All else equal, as price falls the quantity supplied falls and vice versa

# Cont...

## *Supply schedule and curve*

| PRICE OF ICE-CREAM CONE | QUANTITY OF CONES SUPPLIED |
|-------------------------|----------------------------|
| \$0.00                  | 0                          |
| 0.50                    | 0                          |
| 1.00                    | 1                          |
| 1.50                    | 2                          |
| 2.00                    | 3                          |
| 2.50                    | 4                          |
| 3.00                    | 5                          |



# Market Supply Versus Individual Supply Curve

- ➡ Individual Supply Curve – is a curve that represents the price quantity combination of a *particular good* for *a single seller*
- ➡ Market Supply Curve – is a curve that represents the price quantity combination of a particular good for *all sellers*

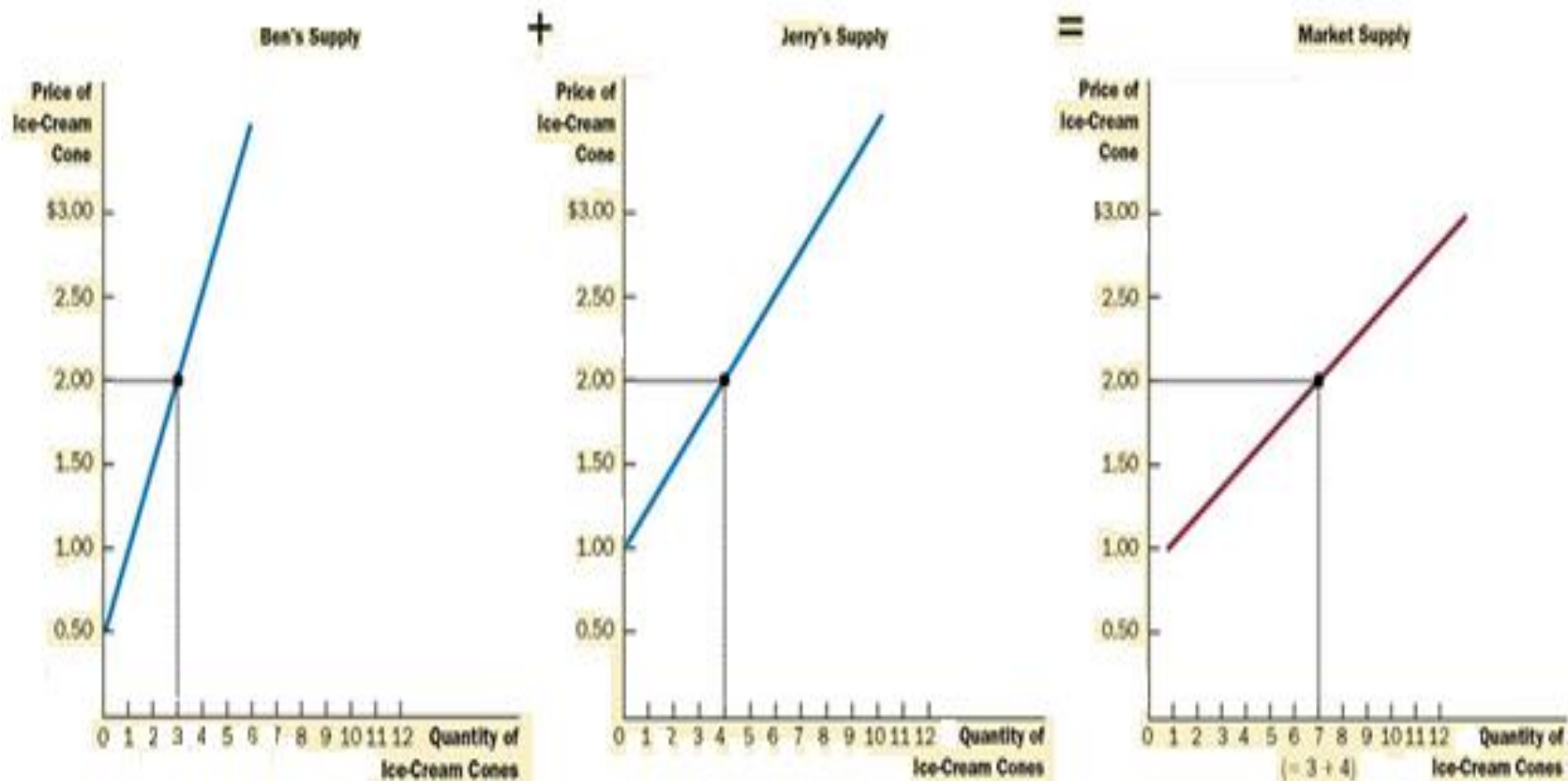
# Cont...

Individual and market supply schedule

| <u>PRICE OF ICE-CREAM CONE</u> | <u>BEN</u> |   | <u>JERRY</u> |   | <u>MARKET</u> |
|--------------------------------|------------|---|--------------|---|---------------|
| \$0.00                         | 0          | + | 0            | = | 0             |
| 0.50                           | 0          |   | 0            |   | 0             |
| 1.00                           | 1          |   | 0            |   | 1             |
| 1.50                           | 2          |   | 2            |   | 4             |
| 2.00                           | 3          |   | 4            |   | 7             |
| 2.50                           | 4          |   | 6            |   | 10            |
| 3.00                           | 5          |   | 8            |   | 13            |

# Cont...

## Individual and market supply curve





# Changes in Quantity Supply vs. Changes in Supply

- Changes in Quantity Supply –is represented by a movement along a given Supply curve  
=> This is caused by the change in the price of the good
- Changes in Supply –is represented by a shift of the Supply curve  
=> This is caused by the change in the determinants of Supply other than price of the product

# Determinants of Supply

1. Price of the product itself
2. Cost of Production
  - a. Prices of required inputs
  - b. Technologies used in production
2. Price of Related Products
3. Taxes and Subsidies
4. Sellers Expectation
5. Number of sellers
6. Weather condition

# Cont...

## ***1. Cost of Production***

### **a. Prices of required inputs**

↑Price of labor

=>hire less people

=>produce less at current P

=>↓S

### **b. Technologies used in production**

New technology

=> produce output for less

=> higher profit on output

=> produce more at current P

=> ↑S

# Cont...

## ***2. Price of Related Products***

Examples: *wheat Vs teff, leather Jacket Vs leather shoe*

↑Price of leather Jacket

=> produce less leather shoe

=> Produce more leather Jacket

=> ↓S of leather shoe

## ***3. Number of sellers***

↑number of sellers

=> More of the good is produced

=> ↑S

## ***4. Expectations***

↑Price tomorrow

=> Sell tomorrow instead of today

=> ↓S

# Cont...

## ***5. Taxes and subsidies***

↑ tax

=> Less of the good is produced

=> ↓S

↑***subsidy***

=> more of the good is produced

⇒↑S

## ***6. Weather condition(Agricultural Products)***

✓ *Good Vs bad weather condition*

Good Weather condition

=> ↑ production of the good

=> ↑S

# Market Equilibrium

🍷 Market Equilibrium – occurs at a point at which the supply and demand curves intersect.

=> occurs when there is no incentive for prices to change (a steady state).

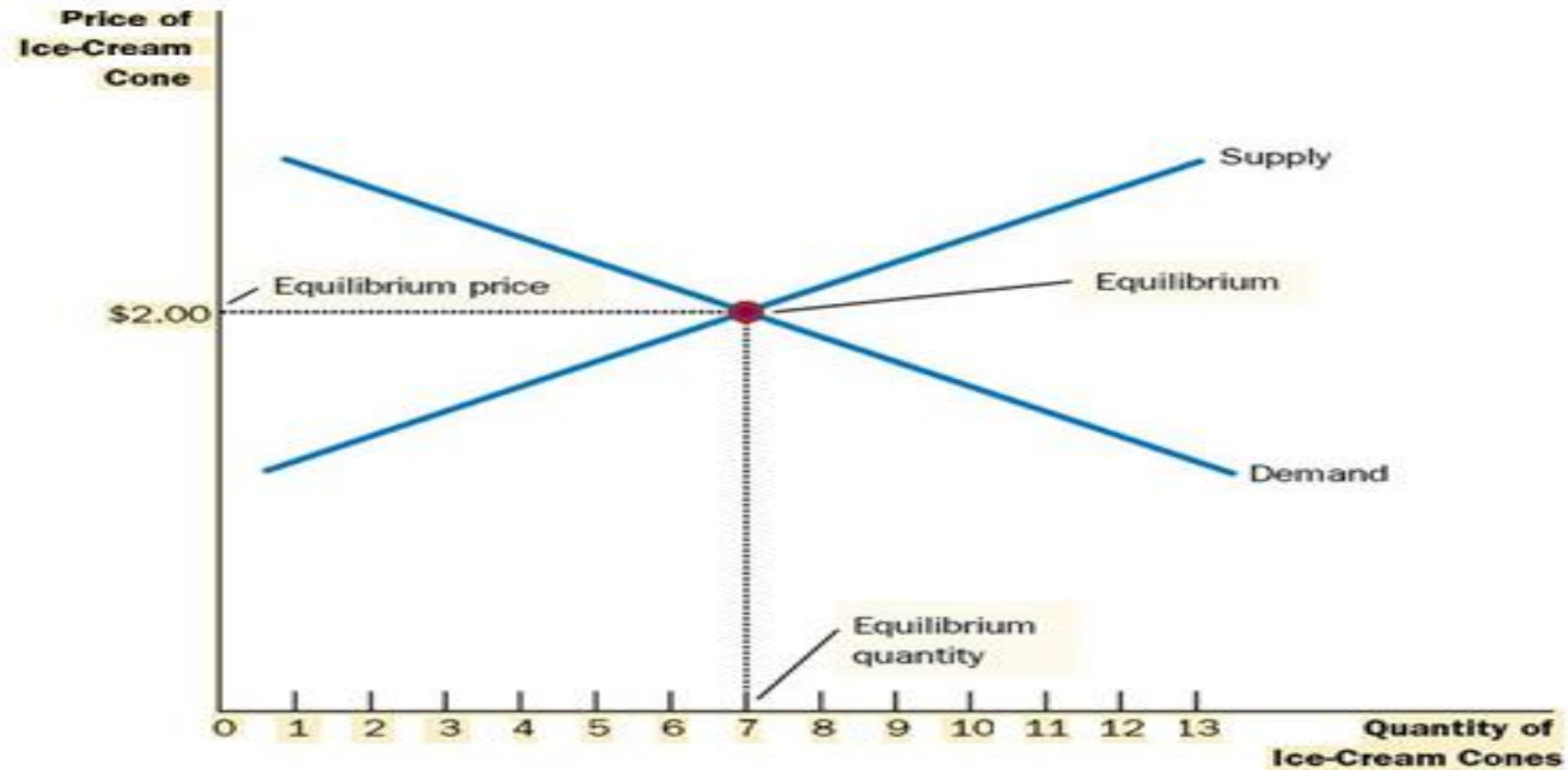
⇒ This occurs when  $Q_S = Q_D$

🍷 The price at which these two curves cross is called the **equilibrium price**

🍷 The quantity at which these two curves cross is called the **equilibrium quantity**

# Cont...

## Market equilibrium



## Cont...

- **Exercise 1:** From statistical studies, we know that for 1981 the supply curve for wheat was approximately as follows:

$$\text{Supply: } Q_S = 1800 + 240P$$

Where price is measured in dollars per bushel and quantities are in millions of bushels per year. These studies also indicate that in 1981 the demand curve for wheat was

$$\text{Demand: } Q_D = 3550 - 266P$$

Find the market clearing price and equilibrium quantity of wheat for the year 1981.

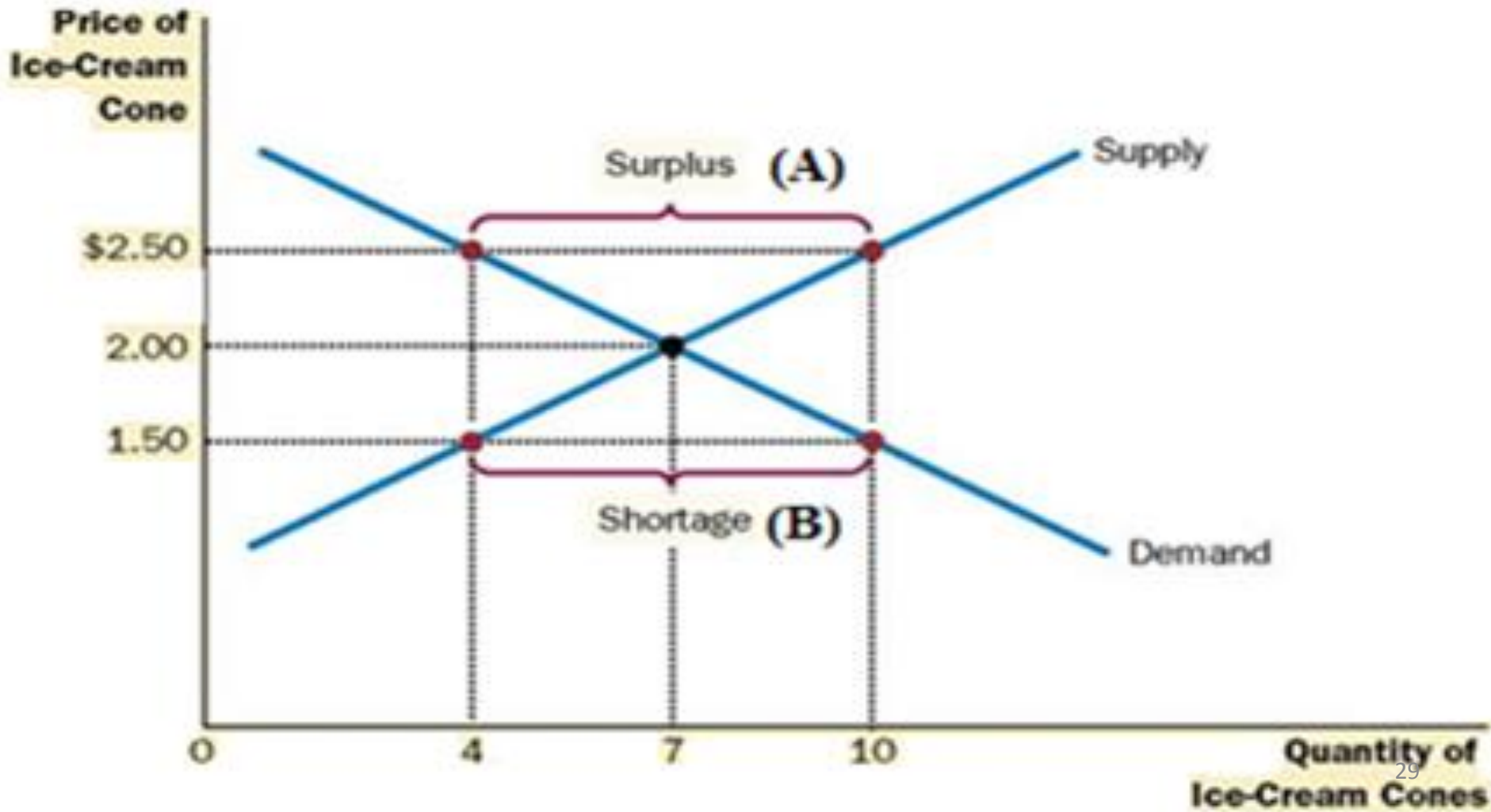


# Cont...

- ❖ Shortage (Excess Demand) – a shortage occurs when the quantity demanded is greater than the quantity supplied at a particular price.
- ❖ Surplus (Excess Supply) – a surplus occurs when the quantity demanded is less than the quantity supplied at a particular price.

# Cont...

Figure for shortage and surplus

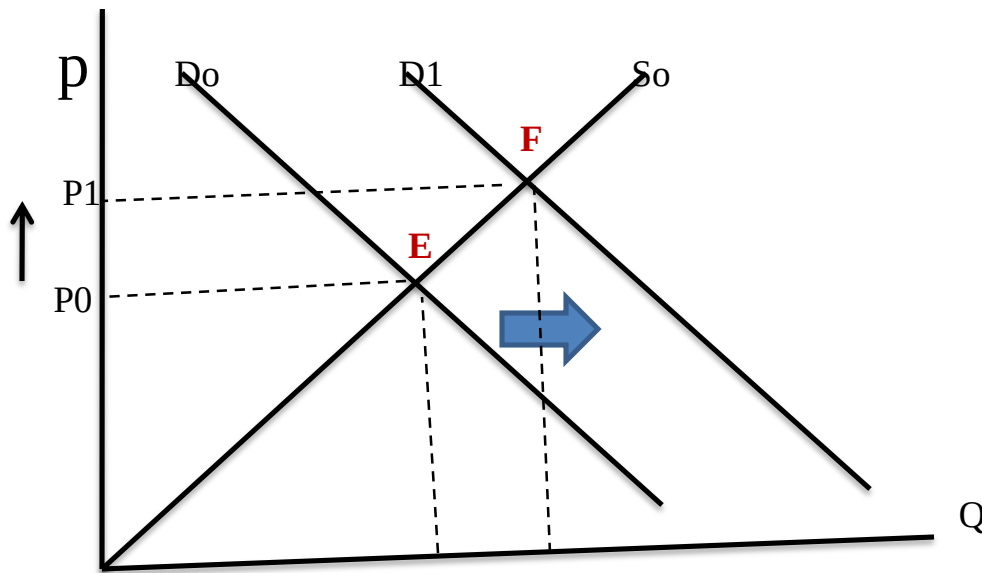


# Effects of changes in demand and supply in equilibrium price and equilibrium quantity

- ➡ Events that affect demand or supply (or both) will alter the equilibrium price and quantity. Let's see for the following conditions;
  - a. Change in demand keeping supply constant
  - b. Change in supply keeping demand constant
  - c. Simultaneous change in demand and supply

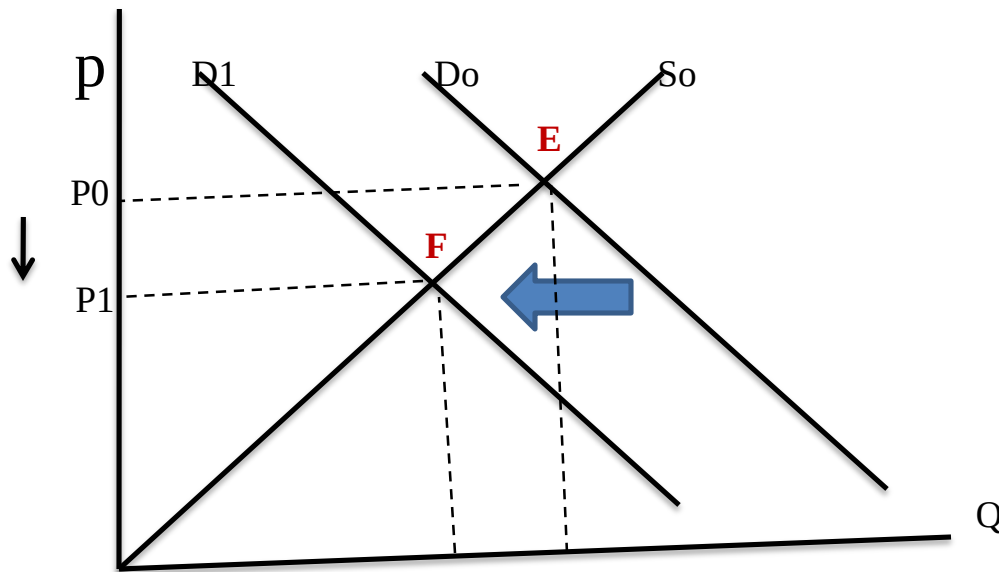
***a. Change in demand keeping supply constant***  
***I. when demand increases***

- + Demand curve shifts to the right
- + Equilibrium point changes
- + Equilibrium price and quantity increases



## *II. when demand decreases*

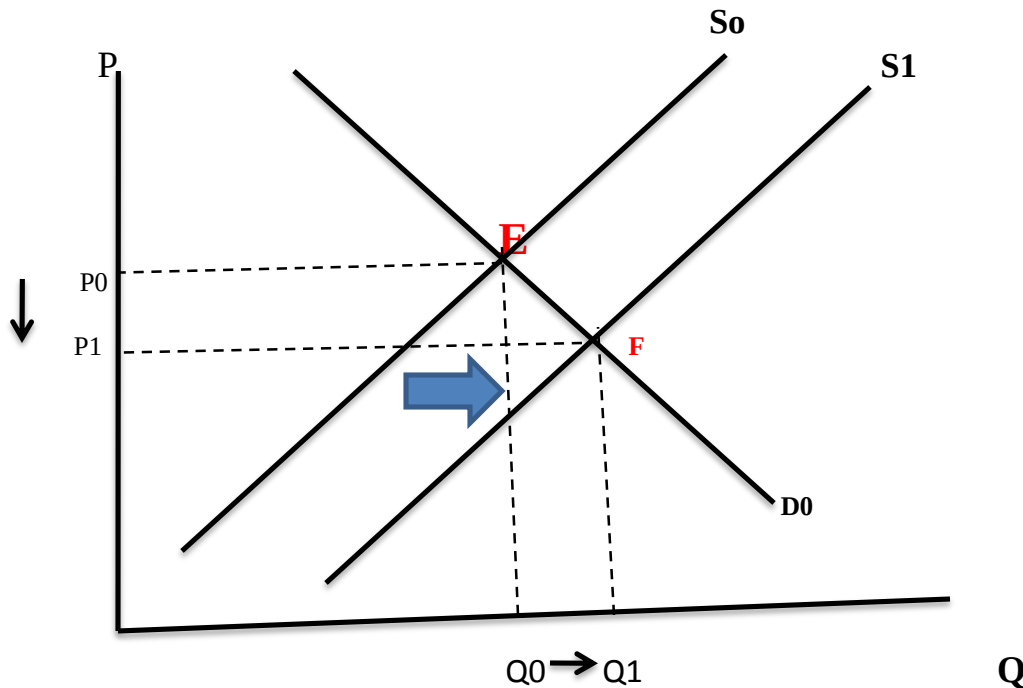
- + Demand curve shifts to the left
- + Equilibrium point changes
- + Equilibrium price and quantity decreases



## ***b. Change in supply keeping demand constant***

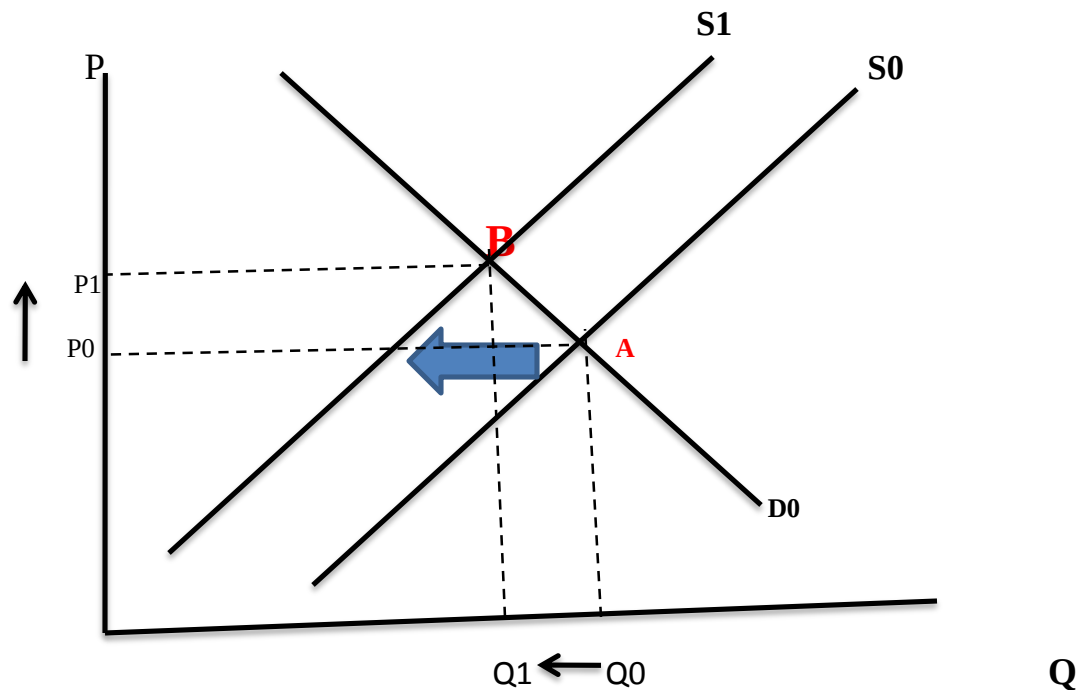
### ***I. when supply increases***

- + Supply curve shifts to the right/outward
- + Equilibrium price decreases but quantity increases



## *II. when supply decreases*

- ✚ Supply curve shifts to the left/inward
- ✚ Equilibrium price increases but quantity decreases



## *c. Change in both demand and supply*

I. When demand increases and supply decreases

When  $DD \uparrow \rightarrow \uparrow P^* \text{ and } Q^* \uparrow$

When  $SS \downarrow \rightarrow \uparrow P^* \text{ and } Q^* \downarrow$

---

**$\uparrow$**   $P^*$ , but  $Q^*$  remains indeterminate



## *Cont...*

II. When demand decreases and supply increases

When DD  $\downarrow \rightarrow \uparrow P^*$  and  $Q^* \downarrow$

When SS  $\uparrow \rightarrow \downarrow P^*$  and  $Q^* \uparrow$

---

$\downarrow P^*$ , but  $Q^*$  remains indeterminate

## *Cont...*

III. When both demand and supply increase

When  $DD \uparrow \rightarrow \uparrow P^* \text{ and } Q^* \uparrow$

When  $SS \uparrow \rightarrow \downarrow P^* \text{ and } Q^* \uparrow$

---

$P^*$  remains indeterminate,  $Q^* \uparrow$

IV. When both demand and supply decrease

When  $DD \downarrow \rightarrow \downarrow P^* \text{ and } Q^* \downarrow$

When  $SS \downarrow \rightarrow \uparrow P^* \text{ and } Q^* \downarrow$

---

$P^*$  remains indeterminate,  $Q^* \downarrow$

# Elasticity Of Demand and Supply

Elasticity refers to responsiveness

**1. Price Elasticity of Demand** - measures how responsive consumers react to price change

Price elasticity of demand =  
$$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

**Point price elasticity of demand**

$$E_D = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

**Arc price elasticity of demand**

$$E_D = \frac{\Delta Q}{\Delta P} \cdot \frac{P+P'}{Q+Q'}$$

# Cont...

■ Elasticity expresses a relationship between two amounts

=> The percent change in quantity demanded

=> The percent change in price

■ The law of demand states that price and quantity demanded are inversely related,

=> the change in price and the change in quantity demanded have opposite signs

=> the price elasticity of demand has a negative sign

- Referring a negative number gets cumbersome, the price elasticity of demand is represented as an absolute value => positive number

# *Categories of Elasticity of Demand*

## **1. Inelastic:**

- ✓ Elasticity is between 0 and 1.0
- ✓ The percent change in quantity demanded is smaller than the percent change in price,
- ✓ Quantity demanded is relatively unresponsive to a change in price

## **2.unit-elastic :**

- ✓ elasticity with an absolute value of 1.0
- ✓ If the percent change in quantity demanded equals the percent change in price

# Cont...

## 3. Elastic:

- ✓ price elasticity has an absolute value exceeding 1.0
- ✓ The percent change in quantity demanded exceeds the percent change in price

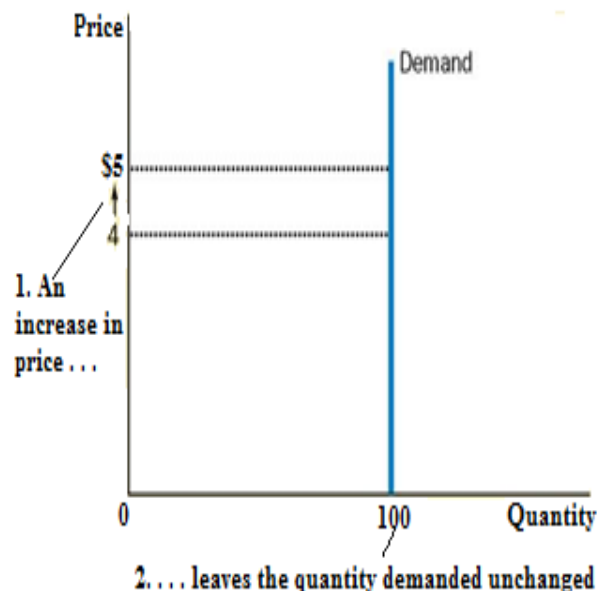
## 4. Perfectly inelastic

- ✓ demand curve is vertical
- ✓ regardless of the price, the quantity demanded stays the same
- ✓ price elasticity of demand approaches zero

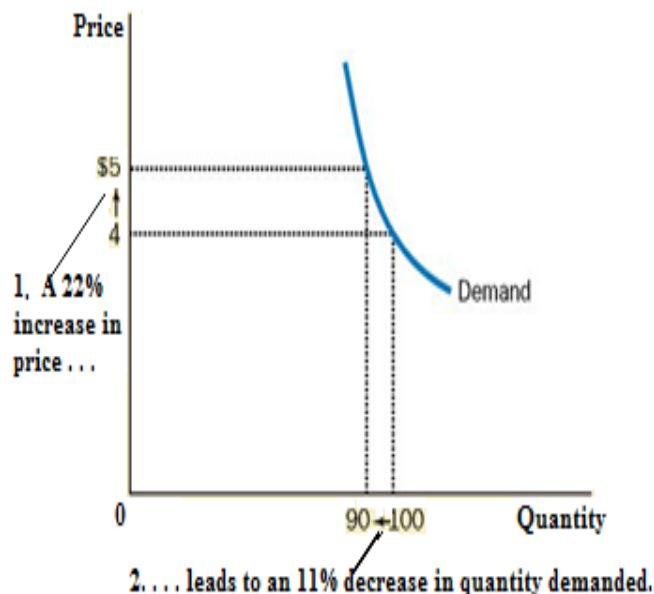
## 5. Perfectly elastic demand

- ✓ price elasticity of demand approaches infinity
- ✓ the demand curve becomes horizontal
- ✓ reflecting the fact that very small changes in the price lead to huge changes in the quantity demanded

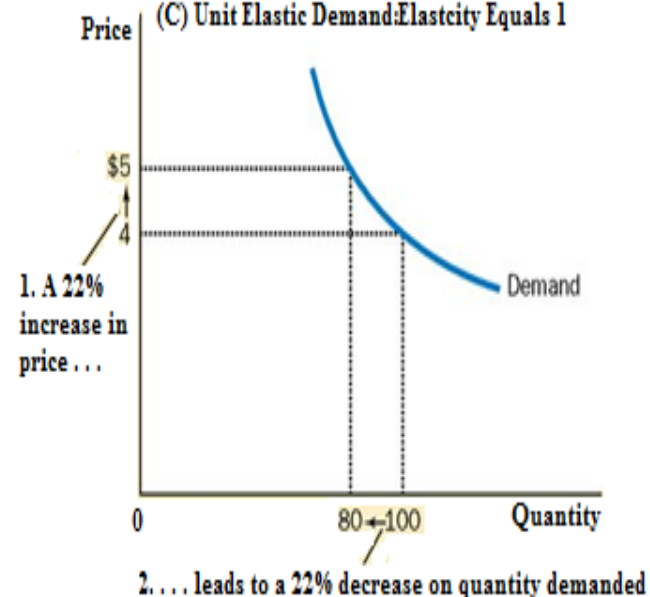
(a) Perfectly Inelastic Demand: Elasticity equals 0



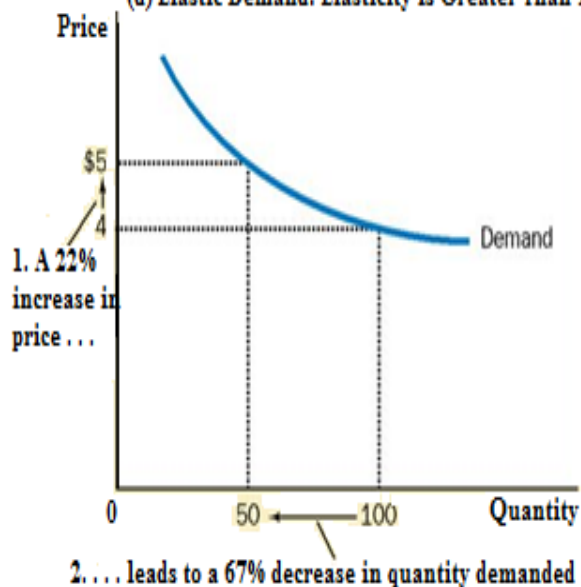
(b) Inelastic Demand: Elasticity Is Less Than 1



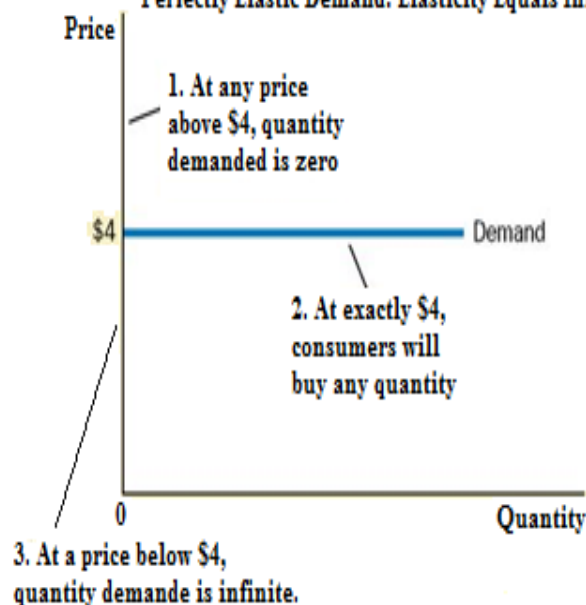
(c) Unit Elastic Demand: Elasticity Equals 1



(d) Elastic Demand: Elasticity is Greater Than 1



Perfectly Elastic Demand: Elasticity Equals Infinity



# *Elasticity and Total Revenue*

- Price elasticity can indicate the effect of a price change on total revenue
- Total revenue (TR) is the price (p) multiplied by the quantity demanded (q) at that price

$$TR = p \times q$$



# Cont...

## ***Relation between Elasticity and Total Revenue***

**When demand is elastic,**

- percent increase in quantity demanded  $\geq$  percent decrease in price
- Total revenue increases

**When demand is unit elastic,**

- percent increase in quantity demanded = percent decrease in price
- Total revenue remains unchanged

**When demand is inelastic,**

- percent increase in quantity demanded  $\leq$  the percent decrease in price
- Total revenue decreases

Cont...

## 2. *Income Elasticity of Demand*

■ Measures the percent change in demand divided by the percent change in income

$$E_I = \frac{\Delta Q}{Q} \cdot \frac{I}{\Delta I}$$

■ Categories of Income Elasticity of Demand

✓ Goods with income elasticity less than zero are called ***inferior goods*** => demand declines when income increases

***Normal goods*** have income elasticity greater than zero => demand increases when income increases

✓ Normal goods with income elasticity greater than zero but less than 1 are called income inelastic goods(***necessary goods***) => demand increases not as much as income does

✓ Goods with income elasticity greater than 1 are called income elastic(***luxury goods***) => demand increases more than does income does

Cont...

### 3. *Cross price elasticity of demand*

■ Measures the percent change in demand of a good divided by the percent change in price of an other good.

$$E_X = \frac{\Delta Q_X}{\Delta P_Y} \cdot \frac{P_Y}{Q_X}$$

■ Categories of cross price Elasticity of Demand

- ✓ Goods with cross price elasticity less than zero are called ***complementary goods***=> demand of a good declines when price of another good increases
- ✓ Goods with cross price elasticity greater than zero are called ***substitute goods***=> demand of a good increases when price of another good increases
- ✓ Goods with cross price elasticity equals to zero are called ***unrelated goods***=> demand of a good doesn't change when price of another good increases

# Cont...

## 4. *Price elasticity of supply*

 Measures how responsive producers are to a price change

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$E_s = \frac{\% \Delta Q_s}{\% \Delta P}$$

### **Point price elasticity of supply**

$$E_s = \frac{\Delta Q_s}{\Delta P} \cdot \frac{P}{Q_s}$$

### **Arc price elasticity of supply**

$$E_s = \frac{\Delta Q_s}{\Delta P} \cdot \frac{P+P'}{Q_s+Q_s'}$$

# *Categories of Supply Elasticity*

 The terminology for supply elasticity is the same as for demand elasticity

- If supply elasticity is less than 1.0, supply is ***inelastic***
- If it equals 1.0, supply is ***unit elastic***
- If it exceeds 1.0, supply is ***elastic***
- If it approaches to infinity, supply is ***Perfectly elastic***
- If it approaches to Zero, supply is ***Perfectly inelastic***

***Exercise 2:*** Plot the graphs of perfectly inelastic and perfectly elastic supply curves.

***Exercise 3:*** Find the price elasticity of demand and supply of wheat for the year 1981 at the equilibrium price and quantity.

***Exercise 4:*** If demand function is given by  $P=50-0.1Q$ . Find the unit elastic point

# CHAPTER THREE

## THEORY OF CONSUMER BEHAVIOR

# Meaning of Utility

Assumptions about the average consumer

1. An average consumer is rational
  - ✓ a consumer has *clear cut* preference – the consumer is able to compare any two bundles ‘a’ and ‘b’ and decide which one he/she prefer
  - ✓ A consumer has *persistent* preference- if the consumer prefers bundle ‘a’ to ‘b’ and ‘b’ to ‘c’, then she/he prefers ‘a’ to ‘c’.
2. Consumer is not free in his/her choice-the consumer choice is limited by his/her *income* level and the *price* of g+s

## Cont'd.....

**Utility** is a pleasure/satisfaction that the consumer obtain by consuming a product.

**It** is a power of product to satisfy human wants.  
In strict sense it doesn't mean realized satisfaction rather expected satisfaction.

- utility is subjective- the utility that two individual derive from consumption of the same level of product is not the same.



# *Theories of utility*

🍷 We have two theories of utility

- ✓ Cardinal utility approach- utility is measurable by arbitrary unit of measurement called utils and it is not comparable
- ✓ Ordinal utility approach- utility is not measurable but comparable

## *1. Cardinal utility approach*

**Total utility** –is total satisfaction a consumer derives from consuming a *specific quantity* of a product at particular time

**Marginal utility**- is extra satisfaction a consumer derives from consuming one more unit of a product. It is the level of utility obtained from the consumption of each additional consumption of g+s.

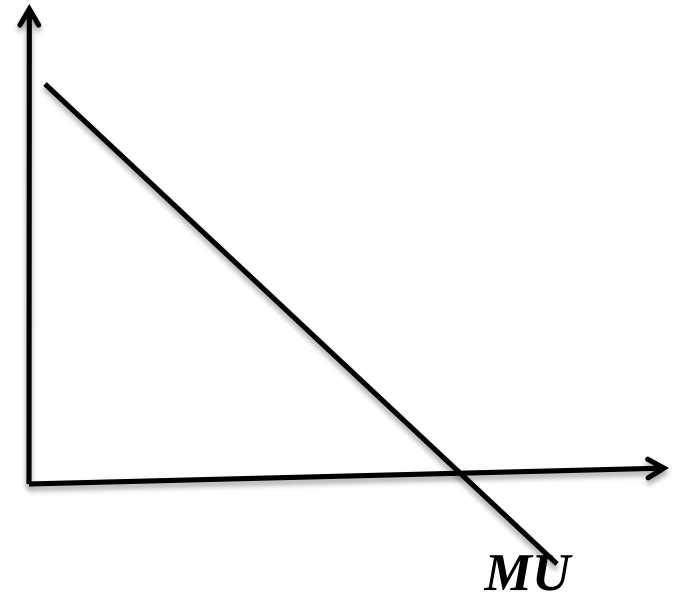
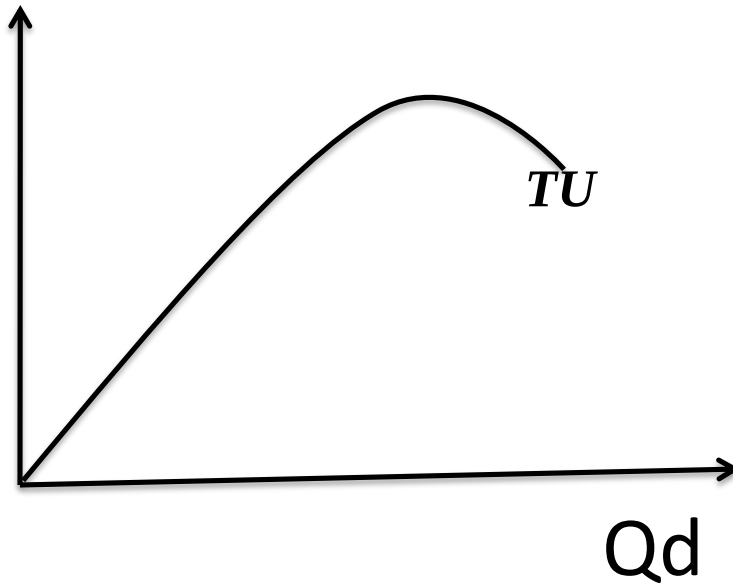
$$Mu = dTu/dQ = \Delta Tu / \Delta Q$$

# ***Numerical example for Tu&Mu***

| <b>Orange consumption</b> | <b>Total utility</b> | <i>Marginal utility</i> |
|---------------------------|----------------------|-------------------------|
| <b>0</b>                  | <b>0</b>             | <b>-</b>                |
| <b>1</b>                  | <b>10</b>            | <b>10</b>               |
| <b>2</b>                  | <b>18</b>            | <b>8</b>                |
| <b>3</b>                  | <b>24</b>            | <b>6</b>                |
| <b>4</b>                  | <b>28</b>            | <b>4</b>                |
| <b>5</b>                  | <b>30</b>            | <b>2</b>                |
| <b>6</b>                  | <b>30</b>            | <b>0</b>                |
| <b>7</b>                  | <b>28</b>            | <b>-2</b>               |

# Cont...

Fig. Tu and Mu



## Cont...

❖ *From the above table and graph, we can deduce the following points*

- 🍊  $Tu$  ↑ first, reaches maximum (when consumers consume 6 units of orange) and then declines as quantity consumed increases
- 🍊  $Mu$  continuously declines (even become zero and negative as quantity consumed increases)

# Law of Diminishing Marginal Utility

- The law states that other things remain constant, consuming successive units of a product gives a consumer less and less extra satisfaction(Mu)
- or in other word, the MU of a goods tends to diminish as the consumer increases the consumption of a good beyond a certain level.

## Assumptions for ***LDMU***

- ✓ The consumer is rational
- ✓ identical product is consumed
- ✓ there is no time gap
- ✓ taste remain unchanged

# *Maximizing utility*

- *Objective of a consumer is total utility maximization*
- *this happens when the last birr spent on each good yield equal MU.*

*Mathematically, the condition of maximum satisfaction and income allocation is as follows*

i.  $\frac{M_{ux}}{P_x} = \frac{M_{uy}}{P_y} = \frac{M_{uz}}{P_z}$

ii.  $P_x.Q_x + P_y.Q_y + P_z.Q_z = M$

*where M is income of the consumer*

## Cont...

Consider that Eyasu has birr 10 and he consumes two goods, X and Y. The price of good X( $P_x$ ) is birr 1 and the price of good Y( $P_y$ ) is birr 2. The total utility he gets from consumption of the goods are given below.

Calculate

- i. Fill the  $M_u$  and  $M_u/P$  for good x and good y
- ii. Find the best combination of x and y that maximizes utility
- iii. Find the maximum utility

| Good X .....Px=birr 1 |     |     |        | Good Y.....Py=birr 2 |     |     |        |
|-----------------------|-----|-----|--------|----------------------|-----|-----|--------|
| Quantity              | TUx | Mux | MUx/Px | Quantity             | TUy | Muy | Muy/Py |
| 0                     | 0   | -   | -      | 0                    | 0   | -   | -      |
| 1                     | 10  | 10  | 10     | 1                    | 24  | 24  | 12     |
| 2                     | 18  | 8   | 8      | 2                    | 44  | 20  | 10     |
| 3                     | 25  | 7   | 7      | 3                    | 62  | 18  | 9      |
| 4                     | 31  | 6   | 6      | 4                    | 78  | 16  | 8      |
| 5                     | 36  | 5   | 5      | 5                    | 90  | 12  | 6      |
| 6                     | 40  | 4   | 4      | 6                    | 96  | 6   | 3      |
| 7                     | 43  | 3   | 3      | 7                    | 100 | 4   | 2      |



**Exercise:** Consider that Molla has birr 12 and he consumes two goods X and Y. The  $P_x=2$  birr and  $P_y= 1$  birr. The TU he gets from the consumption of the goods are given below

- i. Fill the  $M_u$  and  $M_u/P$  for good x and good y
- ii. Find the best combination of x and y that maximizes utility
- iii. Find the maximum utility

# Cont...

| Good X .....Px=birr 2 |     |     |        | Good Y.....Py=birr 1 |     |     |        |
|-----------------------|-----|-----|--------|----------------------|-----|-----|--------|
| Qty                   | Tux | Mux | Mux/Px | Qty                  | Tuy | Muy | Muy/Py |
| 0                     | 0   |     |        | 0                    | 0   |     |        |
| 1                     | 16  |     |        | 1                    | 11  |     |        |
| 2                     | 30  |     |        | 2                    | 21  |     |        |
| 3                     | 42  |     |        | 3                    | 30  |     |        |
| 4                     | 52  |     |        | 4                    | 38  |     |        |
| 5                     | 60  |     |        | 5                    | 45  |     |        |
| 6                     | 66  |     |        | 6                    | 51  |     |        |
| 7                     | 70  |     |        | 7                    | 56  |     |        |
| 8                     | 72  |     |        | 8                    | 60  |     |        |

# Cont...

## *2.Ordinal utility approach/Theory*

- The ordinal theory suggests that utility is only **relatively** discernible but **not** quantifiable, not measurable
- Utility can only be ranked by an order or **a scale of preference** to show the degree of willingness of a consumer
- Since it uses indifference curves to study the consumer behavior, the ordinal utility theory is also known as the **indifference curve approach**.

# Assumptions of IC

1. Consumer is rational
2. Consumer can simply order their preference or they only rank the utility level
3. There is a diminishing marginal rate of substitution- the rate at which one good substitute for another in consumer's basket of goods diminish as the consumer consumes more and more of the goods.
4. The TU of a consumer measured by the amount of all items she/he consumed from his consumption basket.
5. Consumers preference for items in his/her consumption bundle is consistent.....axioms of transitivity.

# The Indifference Set, Curve and Map

## *The Indifference Set*

- ✓ is a combination of goods for which the consumer is indifferent
- ✓ it shows the various combination of goods from which the consumer derives the same level of utility.

| combination | X  | Y |
|-------------|----|---|
| A           | 10 | 2 |
| B           | 6  | 4 |
| C           | 3  | 6 |
| D           | 2  | 8 |

Each combination of good X and Y gives the consumer equal level of utility. Thus the consumer is indifferent whether she/he consumes combination A, B, C or D

# Cont...

## *Indifference Curve*

- ✓ graphical expression combination of goods for which the consumer equal level of utility

## *Indifference Map*

- ✓ is a set of indifference curves
- ✓ Each successive curve to the right represents higher level of utility.

# Characteristics of indifference curves

1. The slope of indifference curve is ***negative*** : there is some degree of substitution between the two goods
  - ❖ As amount of one good increase, the amount of the other good should decrease in order to get the same level of utility.
2. The curve is ***convex to the origin***, i.e. the marginal rate of substitution of two goods is diminishing.

## Cont...

3. A curve further away from the origin means that it stands for a higher level of preference than the one near to the origin
4. There is **no** intersection for any two curves in the indifference map
  - ❖ That is two indifference curves never cross each other.

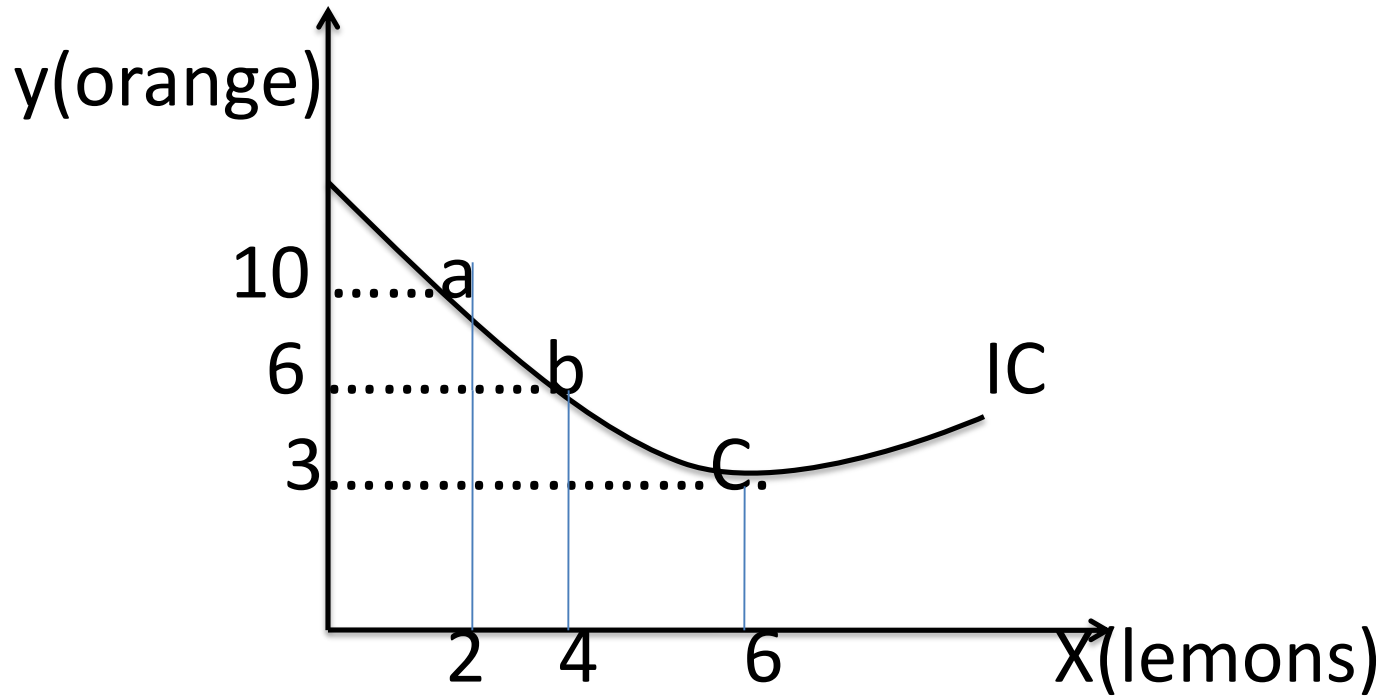


# Marginal rate of substitution(MRS)

MRS – is rate by which the consumer is willing to give up a good so as to obtain more of another good, holding total utility constant

➤  $MRS_{xy} = \Delta Y / \Delta X$

➤ The number of a good Y that had to be given up for each unit of good X to maintain the same level of utility declines, along the indifference curve.



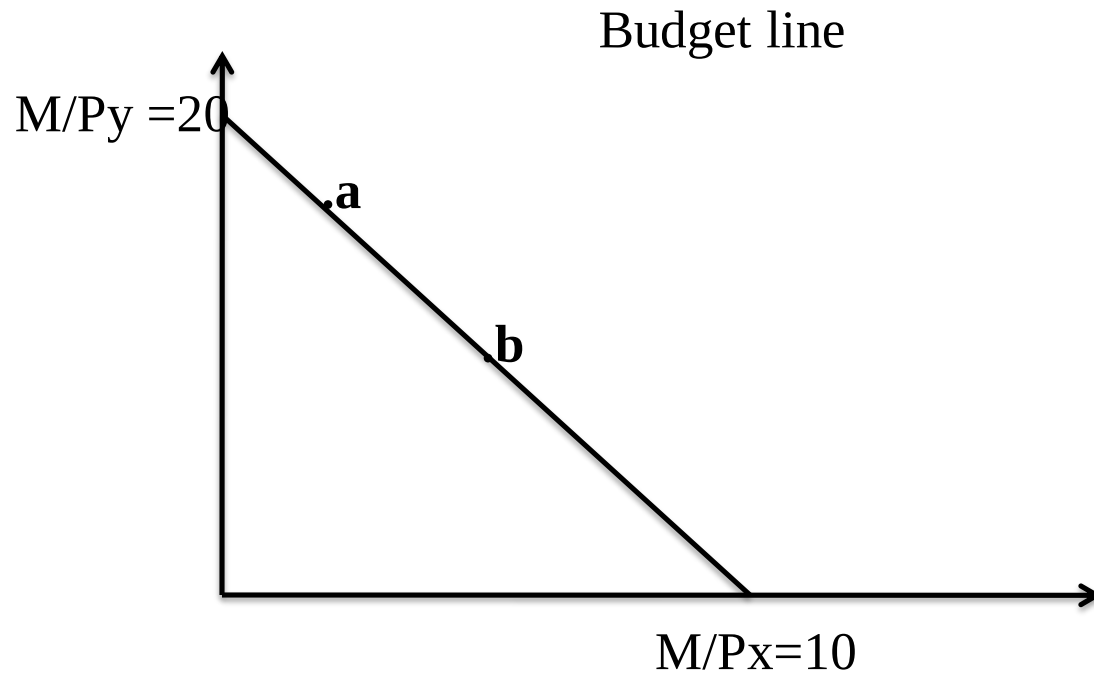
How many of good  $y$ (orange) the consumer is willing to sacrifice to get 2 more unit of good  $x$ (lemons).....4

# Budget line

***Budget line*** - graph that shows the various combinations of two goods that a consumer can buy given his limited income and prices of goods

- The slope of the line gives the relative price of any one good.
- It tells what the consumer must give up in terms of another good in order to buy one good

# Cont...



# Budget line.....

- Consider that dawit has a money income of birr 100, and he consume two goods X and Y. The price of good x is birr 10 and the price of good y is birr 5. If he spend all his income on good y, he would buy 20 units of good y and if he spend all his income on good x, he would buy 10 units of good x.

$$P_x \cdot X + P_y \cdot Y = M$$

- Any combination of two goods on or within the budget line are attainable, and
- Any combination of the two goods above the budget line are unattainable to the consumer, b/c s/he can't afford to buy them.

## Cont...

➤ The slope of the budget line =  $-P_X / P_Y = \Delta Y / \Delta X$   
for the above example the slope is -2.

Therefore, the slope of budget line is the ratio of the prices of the two goods.

It shows the market willingness not consumer willingness to substitute one good for another.

### ***Factors which affect budget line***

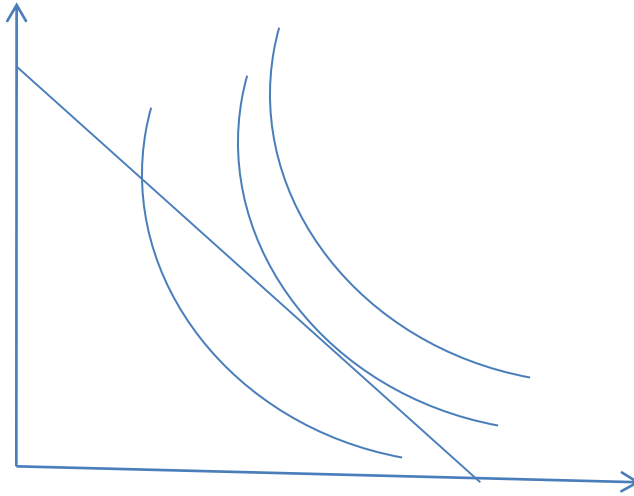
1. Income (an increase and decrease in income)
2. Change in price (of one good or both)

# Maximizing Total Utility

- A consumer gets the maximum possible utility when he/she buys that combination of goods at which the budget line is tangent to the highest attainable indifference curve
- An equilibrium point or satisfaction unit is
  1. BL and IC are tangent at each other
  2. slope of BL = slope of IC

$$P_X / P_Y = \Delta Y / \Delta X = MU_X / MU_Y$$

# Cont.....





# CHAPTER FOUR

## THEORY OF PRODUCTION AND COST

# Definition of Production and Production Function

## *Production*

- is the process of changing inputs into output
- Is the process of using the services of inputs to make  $g+s$  available.

## *Input*

- is anything that can be used in the production of  $g+s$ . include labor, capital, land and entrepreneurship.

There are two types of inputs

# Cont....

➡ ***Fixed inputs***-inputs that cannot vary overtime

Or inputs whose quantity remain fixed for a given period of time.

Example: machineries, capital

➡ ***Variable inputs*** –those inputs whose quantity vary overtime

- Example: labour

# ***Production Function***

- shows the relationship between inputs and the resulting output
- It describe the transformation of input into output.
- indicates the output  $Q$  that a firm produces for every specified combination of inputs
- For example if we assume that there are only two inputs, labor ( $L$ ) and capital ( $K$ ). We can then write the production function as
  - $Q = F(K, L)$
- The equation relates the quantity of output to the quantities of the two inputs, capital and labor

Production period

## **The Short run vs Long run period**

*Showing the nature of economic adjustment in the firm to changing economic environment.*

### ***Short run***

➡ refers to a period of time in which at least one input remain fixed.

### ***Long run***

➡ refers to a period of time in which all factors of production are variable

# The Short run Production Function

- The production function is said to short run if there exist at least one fixed input in the production.
- In this case it is assumed that there is only one variable input
- This variable input is usually called “*labor*”
- It is also assumed that this variable input can be combined in different proportions with the fixed variable input to produce various quantities of output.

# Cont...

- Consider the case in which capital is fixed, but labor is variable, so that the firm can produce more output by increasing its labor input
- Imagine, for example, that you are managing a clothing plant
- You have to decide how much labor to hire and how much clothing to produce
- To make the decision, you will need to know how the amount of output  $Q$  increases (if at all), as the input of labor  $L$  increases

# Cont...

- The three columns in the following table(table 4.1) gives you this information

## ***Measurements of output***

### ***1.Total products***

- The total amount of output produced

### ***2.Average product***

- The total output produced per unit of variable input(labor)

- $APL=Q/L$

### ***3. Marginal Product***

- is the additional output produced as the labor input is increased by one unit

- $MPL= \Delta Q/\Delta L$



# Table 4.1

| <i>Amount of Labor<br/>(L)</i> | <i>Amount of Capital<br/>(K)</i> | <i>Total Output (Q)</i> | <i>Average Product<br/>(Q/L)</i> | <i>Marginal Product<br/>(<math>\Delta Q/\Delta L</math>)</i> |
|--------------------------------|----------------------------------|-------------------------|----------------------------------|--------------------------------------------------------------|
| <b>0</b>                       | <b>10</b>                        | <b>0</b>                |                                  | <b>-</b>                                                     |
| <b>1</b>                       | <b>10</b>                        | <b>10</b>               | <b>10</b>                        | <b>10</b>                                                    |
| <b>2</b>                       | <b>10</b>                        | <b>30</b>               | <b>15</b>                        | <b>20</b>                                                    |
| <b>3</b>                       | <b>10</b>                        | <b>60</b>               | <b>20</b>                        | <b>30</b>                                                    |
| <b>4</b>                       | <b>10</b>                        | <b>80</b>               | <b>20</b>                        | <b>20</b>                                                    |
| <b>5</b>                       | <b>10</b>                        | <b>95</b>               | <b>19</b>                        | <b>15</b>                                                    |
| <b>6</b>                       | <b>10</b>                        | <b>108</b>              | <b>18</b>                        | <b>13</b>                                                    |
| <b>7</b>                       | <b>10</b>                        | <b>112</b>              | <b>16</b>                        | <b>4</b>                                                     |
| <b>8</b>                       | <b>10</b>                        | <b>112</b>              | <b>14</b>                        | <b>0</b>                                                     |
| <b>9</b>                       | <b>10</b>                        | <b>108</b>              | <b>12</b>                        | <b>-4</b>                                                    |
| <b>10</b>                      | <b>10</b>                        | <b>100</b>              | <b>10</b>                        | <b>-8</b>                                                    |

# ***Stages Of Production***

In SRPF there are 3 stages of production

## ***Stage I***

- Starts from the origin and ends at maximum point of AP( $AP=MP$ )
- TP and AP increases throughout this stage
- MP at first increases then starts to decline
- The amount of variable input is small as compared to the fixed input. i.e. labor is underemployed and hence fixed input is under utilized.

# Cont...

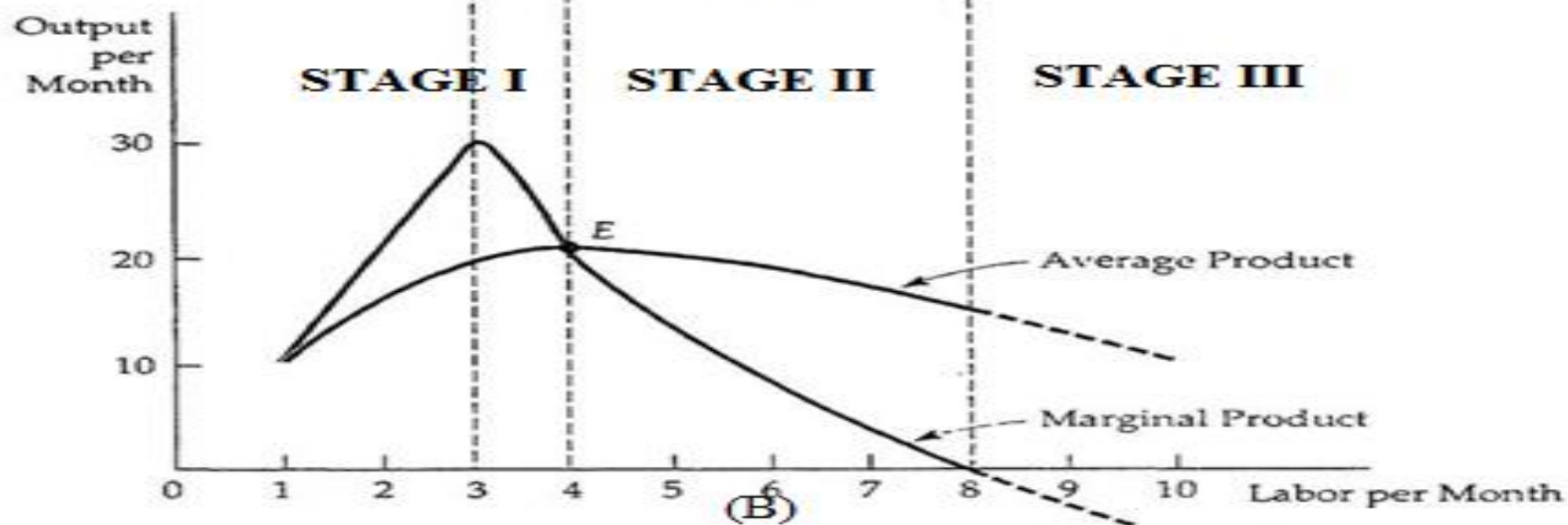
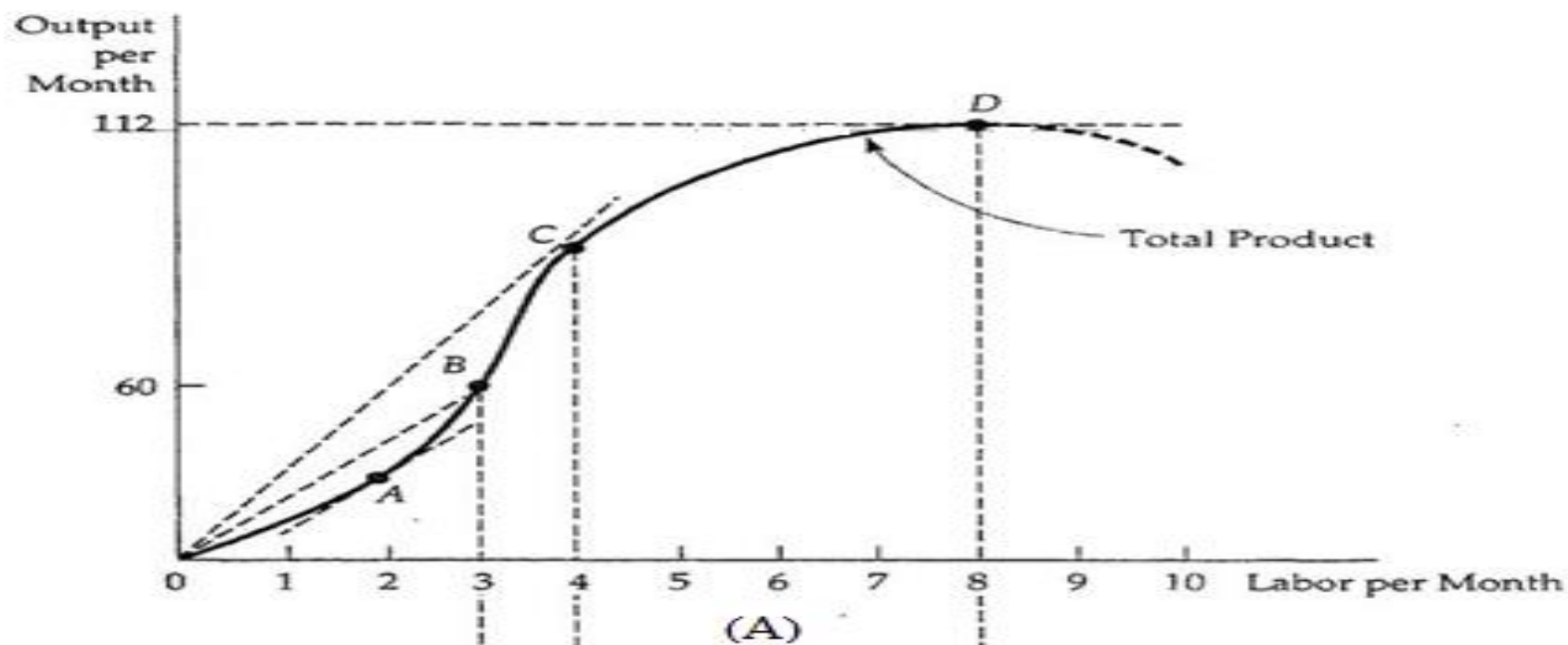
## *Stage II*

- Starts from maximum point of AP( $AP=MP$ ) the and ends at maximum point of TP( $MP = 0$ )
- AP and MP are +ve but decrease throughout this stage
- TP increases but at decreasing rate
- The amount of variable input is proportional with the fixed input
- rational producer should produce at this stage since this stage is the optimum stage.

# Cont...

## *Stage III*

- Starts from the maximum point of TP and endless (whenever MP is negative)
- TP and AP decreases throughout this stage
- MP decreases and even negative
- The amount of variable input is greater as compared to the fixed input
- Each additional unit of labor contributing negatively to total product due to excess of variable input over fixed input. i.e. over utilization due to over employment....due to this, this stage also known as extensive margin.



# The law of diminishing marginal returns

- The law states that as increasing amount of variable input is combined with fixed inputs, eventually the contribution of each additional amount of the variable input to the TP declines.
- The law starts to operate after the MP curve reaches its maximum.

## Cont.....

- When total product is maximum, marginal product is zero
- When  $M_p$  equals  $AP$ ,  $AP$  reaches its maximum.
- whenever,  $MP$  of the variable input is less than  $AP$ , the  $AP$  decreases.
- Whenever  $MP > AP$ ,  $AP$  increases

# Production function with two variable input

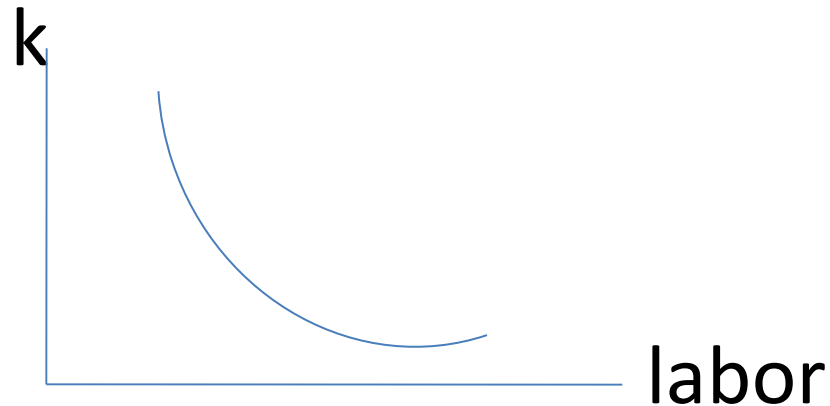
- When Labor and capital are variable input...long run production
- Given two variable input one can produce the same level of product by different combination of those two input.
- A table that represent the various combination of labor and capital which give the producer the same level of output is called *isoquant schedule* or *equal product schedule*.



*isoquant schedule or equal product schedule.*

| Combination | Capital | Labor | Maximum output | MRTs of capital for labor |
|-------------|---------|-------|----------------|---------------------------|
| A           | 1       | 11    | 60             | -                         |
| B           | 2       | 7     | 60             | 4:1                       |
| C           | 3       | 4     | 60             | 3:1                       |
| D           | 4       | 2     | 60             | 2:1                       |
| E           | 5       | 1     | 60             | 1:1                       |

- A graphical representation of isoquant schedule is called iso product curve or equal product curve or simply isoquant.
- Isoquant refers to those combination of two variable inputs that yields the same level of output.



# Cont...

- Isoquant map is the set of isoquant or equal product curves.
- Each successive isoquant to the right represents higher level of total product because it reflect the use of more of at least one of the two input.

## **Properties of isoquant**

1. Isoquant slopes downward
2. It is concave to the origin.....MRTS
3. Isoquant never cross each other

# The economic region of production

- the the region in which the producer should produce in case of two variable input like in the case of stage II for short run prodction.
- Implying that the producer should select those combinations that may not employee him/her more of labor or capital

# Marginal rate of technical substitution (MRTS)

Shows an entrepreneurs willingness to substitute one input by another input so as to be efficient in production.

The rate at which one input is substituted by another input to get the same level of output is called MRTS.

$$\text{MRTS} = \Delta L / \Delta K$$

*(See the previous example)....show that the principle of diminishing marginal rate of technical substitution, determining the shape of the isoquant*

# Return to scale

- Is a property of production function that indicate the relationship between proportionate change in all inputs and the resulting change in total product.
- It is a property that apply only in the long run.
- Three types
  1. constant return to scale:-  $\% \Delta$  in all input =  $\% \Delta$  in output
  2. Increasing return to scale:-  $\% \Delta$  in all input <  $\% \Delta$  in output
  3. Decreasing return to scale :-  $\% \Delta$  in all i/pt >  $\% \Delta$  in output

# Increasing return to scale may occur because of:

- a. An increase in the scale of operation...using specialized machinery
- b. Greater division of labor
- c. Greater degree of specialization

## **effect of technological change on production function**

- Technological advancement makes our limited resource more productive.
- More output from the existing level of inputs
- The existing level of output by using less of the inputs
- It shift the TP curve up(isoquant curve down) implying that we can produce more of the product than before.

# THEORY OF COST

## **Cost**

- Is the monetary value of inputs used in production of  $g+s$
- two types
  1. ***social costs***- *is the cost of producing an item to the society.*
  2. ***private costs***- *is the cost of producing an item to the individual producer.*

Private costs can be measured in two ways

- ❖ ***Accounting cost = explicit cost***(the actual or out of pocket expenditure)
- ❖ ***Economic cost = Explicit cost+ Implicit cost***(the estimated cost of non purchased input)

It is the cost of all inputs



# Analysis of costs in the short-run

- In the short run, some of the firm's inputs to production are fixed, while others can be varied as the firm changes its output
- Various measures of the cost of production can be distinguished on this basis
- ❖ ***Total Cost (TC)***: The total cost of production has two components:
  1. ***Fixed cost (FC)***, which is borne by the firm whatever level of output it produces
  2. ***Variable cost (VC)***, which varies with the level of output

$$TC = TFC + TVC$$

# Cont...

## ❖ *Marginal Cost (MC):*

- ➡ is the increase in cost that results from producing one extra unit of output
- ➡ Marginal cost is just the increase in variable cost that results from an extra unit of output.
- ➡ We can therefore write marginal cost as

$$MC = \Delta TC / \Delta Q = \Delta TVC / \Delta Q$$

# Cont...

## ***Average cost (ATC):***

➡ is the cost per unit of output. Average total cost (ATC) is the firm's total cost divided by its level of output.

$$\text{➡ } ATC = TC/Q$$

**ATC** has two components.

***Average fixed cost (AFC)*** is the fixed cost divided by the level of output.

$$AFC = FC/Q$$

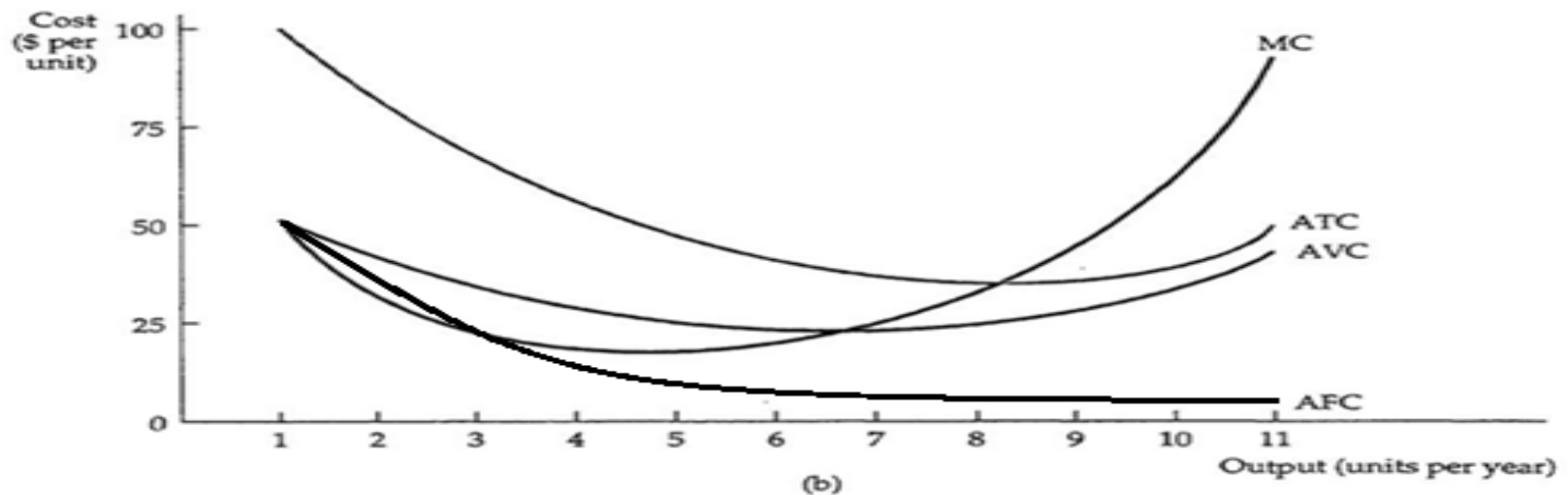
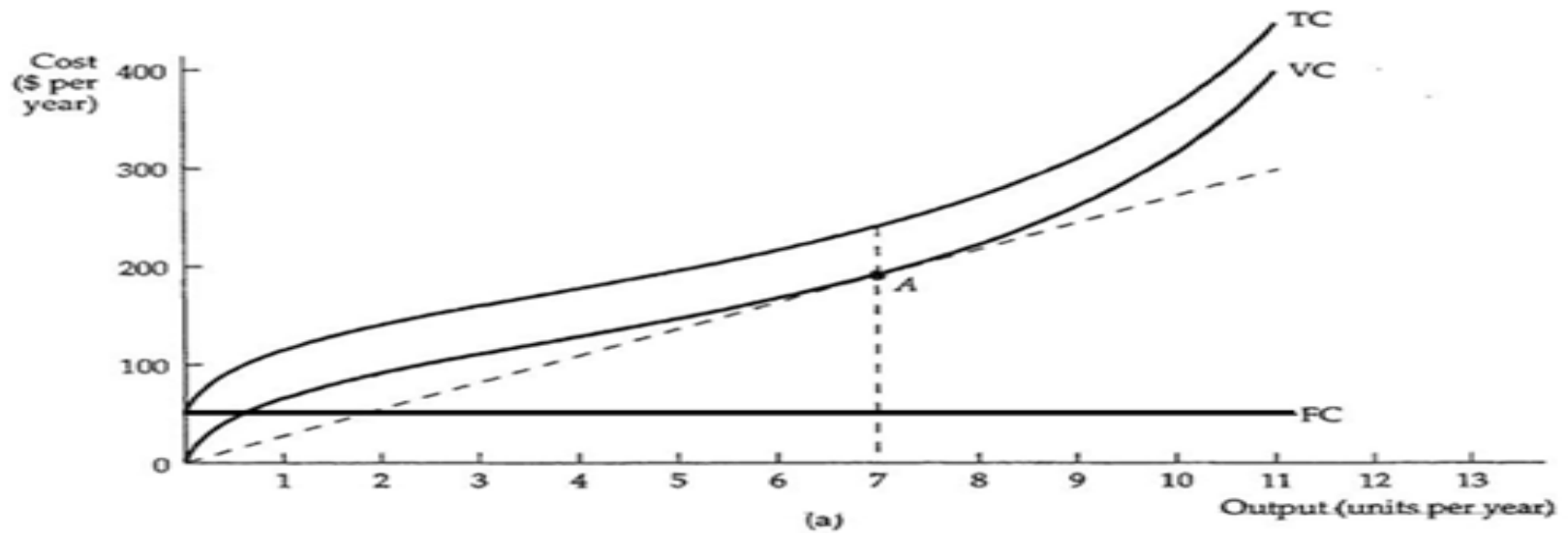
***Average Variable Cost (AVC)***: is variable cost divided by the level of output.

$$AVC = VC/Q$$

# Table for cost measurements

| <b>Rate of Output</b> | <b>Fixed Cost</b> | <b>Variable Cost</b> | <b>Total Cost</b> | <b>Marginal Cost</b> | <b>Average Fixed Cost</b> | <b>Average Variable Cost</b> | <b>Average Total Cost</b> |
|-----------------------|-------------------|----------------------|-------------------|----------------------|---------------------------|------------------------------|---------------------------|
|                       | <b>(FC)</b>       | <b>(VC)</b>          | <b>(TC)</b>       | <b>(MC)</b>          | <b>(AFC)</b>              | <b>(AVC)</b>                 | <b>(ATC)</b>              |
| <b>0</b>              | <b>50</b>         | <b>0</b>             | <b>50</b>         | <b>-</b>             | <b>-</b>                  | <b>-</b>                     | <b>-</b>                  |
| <b>1</b>              | <b>50</b>         | <b>50</b>            | <b>100</b>        | <b>50</b>            | <b>50</b>                 | <b>50</b>                    | <b>100</b>                |
| <b>2</b>              | <b>50</b>         | <b>78</b>            | <b>128</b>        | <b>28</b>            | <b>25</b>                 | <b>39</b>                    | <b>64</b>                 |
| <b>3</b>              | <b>50</b>         | <b>98</b>            | <b>148</b>        | <b>20</b>            | <b>16.7</b>               | <b>32.7</b>                  | <b>49.3</b>               |
| <b>4</b>              | <b>50</b>         | <b>112</b>           | <b>162</b>        | <b>14</b>            | <b>12.5</b>               | <b>28</b>                    | <b>40.5</b>               |
| <b>5</b>              | <b>50</b>         | <b>130</b>           | <b>180</b>        | <b>18</b>            | <b>10</b>                 | <b>26</b>                    | <b>36</b>                 |
| <b>6</b>              | <b>50</b>         | <b>150</b>           | <b>200</b>        | <b>20</b>            | <b>8.3</b>                | <b>25</b>                    | <b>33.3</b>               |
| <b>7</b>              | <b>50</b>         | <b>175</b>           | <b>225</b>        | <b>25</b>            | <b>7.1</b>                | <b>25</b>                    | <b>32.1</b>               |
| <b>8</b>              | <b>50</b>         | <b>204</b>           | <b>254</b>        | <b>29</b>            | <b>6.3</b>                | <b>25.5</b>                  | <b>31.8</b>               |
| <b>9</b>              | <b>50</b>         | <b>242</b>           | <b>292</b>        | <b>38</b>            | <b>5.6</b>                | <b>26.9</b>                  | <b>32.4</b>               |
| <b>10</b>             | <b>50</b>         | <b>300</b>           | <b>350</b>        | <b>58</b>            | <b>5</b>                  | <b>30</b>                    | <b>35</b>                 |
| <b>11</b>             | <b>50</b>         | <b>385</b>           | <b>435</b>        | <b>85</b>            | <b>4.5</b>                | <b>35</b>                    | <b>39.5</b>               |

# Cost curves for a firm



# The Shapes of the Cost Curves(from fig a & b)

- ➡ **Fixed cost  $FC$**  does not vary with output and is shown as a horizontal line at \$50.
- ➡ **Variable cost  $VC$**  is zero when output is zero, and then increases continuously as output increases.
- ➡ **Total cost curve  $TC$**  is determined by vertically adding the fixed cost curve to the variable cost curve.
- ➡ Because fixed cost is constant, the vertical distance between the two curves is always \$50.

## Cont...

- Figure b shows the corresponding set of marginal and average variable cost curve
- Since total fixed cost is \$50, the average fixed cost curve AFC falls continuously from \$50 toward zero
- The shape of the remaining short-run cost curves is determined by the relationship between the marginal and average cost curves

# Cont...

- ➡ Whenever marginal cost lies below average cost, ( $MC < AC$ ) the average cost curve falls.
- ➡ Whenever marginal cost lies above average cost, ( $MC > AC$ ) the average cost curve rises. And when average cost is at a minimum, marginal cost equals average cost
- ➡ The ATC curve shows the average total cost of production
- ➡ The vertical distance between the ATC and AVC curves decreases as output increases



## Cont...

- ➡ The AVC cost curve achieves its minimum point at a lower output than the ATC curve
- ➡ This follows because  $MC = AVC$  at its minimum point, and  $MC = ATC$  at its minimum point.
- ➡ Since ATC is always greater than AVC and the marginal cost curve MC is rising, the minimum point of the ATC curve must lie above and to the right of the minimum point of the AVC curve.

# CHAPTER FIVE

## MARKET STRUCTURE

# Market Structure

- ➡ Market is a mechanism that brings buyer and seller together.
- ➡ A firm's decision concerning price and production depends greatly on the character of the industry in which it is operating
- ➡ Economist group industries into four distinct market structures:
  - ❖ *pure competition(perfectly competitive market)*
  - ❖ *pure monopoly*
  - ❖ *monopolistic competition(monopolistically CM)*
  - ❖ *oligopoly*

# Con'd

- ➡ These four market models differ in several respects:
  - ✓ *The number of firms in the industry*
  - ✓ *Whether those firms produce a standardized product or try to differentiate their products from those of other firms*
  - ✓ *How easy or how difficult it is for firms to enter the industry*

# PURE COMPETITION(PCM)

- ➡ Involves a very large number of buyers and sellers of a product.
- ➡ firms producing a **standardized(homogenous)** product (that is, a product identical to that of other producers, such as corn)
- ➡ Perfect knowlegde or perfect information : every buyer and seller has full information about the market and nature of the product.
- ➡ Free entry and exit: New firms can enter or exit the industry very easily

## PCM....con't

### *Features of PCM structure*

- ✓ A PCM firm is price taker not price setter.
- ✓ It faces a horizontal, or perfectly elastic demand curve
- ✓ The firms total profit obtained through

$$\Pi = TR - TC$$

....Here 3 possibilities.....positive or superior ec/c profit, normal profit or incur loss

- ✓ the firm maximize its profit when

*i. MR=MC*

*ii. MC is rising*

# Imperfect market

- The market with imperfect competition
- It exist when a single firm has a certain degree of control over the market price of a product.(this happen when one or more condition of perfect market is violated.)

3 major types

- ✓ Pure monopoly
- ✓ Oligopoly
- ✓ Monopolistically competitive market

# Pure monopoly

- Is the market structure in which there is only one firms that produces a distinctive product.
  - By distinctive product we mean a product which have no close substitute.
- Only one supplier of a product
  - The product has no close substitute (unique product)
  - There is considerable entry barrier for a new firms...legal and patent right, control over essential raw material, technical, economies of scale, or any other
  - A pure monopoly firm is a price setter, not price taker
  - Profit maximizing condition is  $MR=MC$  and  $MC$  is rising



# Pure Monopoly

- ➡ is a market structure in which one firm is the sole seller of a product or service (for example, a local electric utility).
- ➡ Since the entry of additional firm is blocked, one firm constitutes the entire industry
- ➡ Because the monopolist produces a unique product, it makes no effort to differentiate its product
- ➡ No close substitutes

# Monopolistic competition

- ➡ Is characterized by a relatively large number of sellers producing differentiated products (clothing, furniture, books)
- ➡ The existence of non price competition: There is widespread non price competition, a selling strategy in which one firm tries to distinguish its product or service from all competing products on the basis of attributes like design and workmanship (an approach called product differentiation).
- ➡ Either entry to or exit from monopolistically competitive industries is reasonable.
- ➡ No collusion among the firms
- ➡ Profit maximizing condition is  $MR=MC$  and  $MC$  is rising
- ➡ Like the PCM firm MCM firm earn normal profit in the long run

# Oligopoly

- ➡ involves only a few sellers of an identical or similar product; consequently, each firm is affected by the decisions of its rivals and must take those decisions into account in determining its own price and output.
- ➡ A special type of oligopoly where there are only two firms is called *duopoly*
- ➡ The existence of few dominant firms
- ➡ Firms are mutually interdependent
- ➡ They produce standardize or differentiated product
- ➡ There is non price competition if the product is differentiated product

| <i>Characteristic</i>        | <b>Market Model</b>     |                                                               |                                                                   |                                      |
|------------------------------|-------------------------|---------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------|
|                              | Pure Competition        | Monopolistic Competition                                      | Oligopoly                                                         | Pure Monopoly                        |
| <i>Number of firms</i>       | A very large number     | Many                                                          | Few                                                               | One                                  |
| <i>Type of product</i>       | Standardized            | Differentiated                                                | Standardized or differentiated                                    | Unique; no close substitutes         |
| <i>Control over price</i>    | None                    | Some, but within rather narrow limits                         | Limited by mutual interdependence; considerable with collusion    | Considerable                         |
| <i>Condition of entry</i>    | Very easy, no obstacles | Relatively easy                                               | Significant obstacles                                             | Blocked                              |
| <i>Non price competition</i> | None                    | Considerable emphasis on advertising, brand names, trademarks | Typically a great deal, particularly with product differentiation | Mostly public relations, advertising |
| <i>Examples</i>              | Agriculture             | Retail trade, dresses, shoes                                  | Steel, automobiles, farm implements, many household appliances    | Local utilities                      |

# CHAPTER SIX

FUNDAMENTAL CONCEPTS OF  
MACROECONOMICS AND  
MACROECONOMIC PROBLEMS

# Fundamental Concepts of Macroeconomics

Macroeconomics is a branch of economics which deals with the economy as a whole.

***Macroeconomics concerns are (Macroeconomics concerned with ):***

- ➡ economic growth(aggregate output level)
- ➡ employment
- ➡ inflation (aggregate price level)
- ➡ economic fluctuation
- ➡ distribution of income
- ➡ macroeconomic policies
- ➡ international trade and resource flows(relationship of domestic economy with the rest of the world)

# Cont...

***The objectives of macroeconomics are:***

- ➡ ***Full employment*** – not only providing job for all that are willing and able to work, but also full utilization of other economic resources.
- ➡ ***Price stability***
- ➡ ***Economic growth and rising the living standard of the citizen***
- ➡ ***Fair distribution of income among all citizen of the country***
- ➡ ***Less fluctuation in economic activities***

# Components of macro economy

It focuses on four groups in the economy: *households*, *firms*(the private sector), the *government*(the public sector) and *the rest of the world*(the international sector).

## Problems of Macroeconomics

*The following are the fundamental macroeconomic problems:*

1. Business cycle
2. Unemployment
3. Inflation
4. Budget and trade deficit



# 1. Business Cycle

- ➡ refers to the recurrent ups and downs (fluctuations) in the level of economic activity
- ➡ countries usually experience ups and downs in the level of total output and employment through time
- ➡ There are four phases of in business cycle

## ***1. Boom or Peak***

- ➡ When the economy produce the highest level of output in business cycle
- ➡ It is a period of maximum output expansion and very low level of unemployment
- ➡ The economy is operating close to full capacity
- ➡ Period of prosperity

# Cont...

## ***2. Recession(contraction)***

- ➡ the level of economic performance declines
- ➡ output, national income and employment declines
- ➡ Unemployment problem began to rise

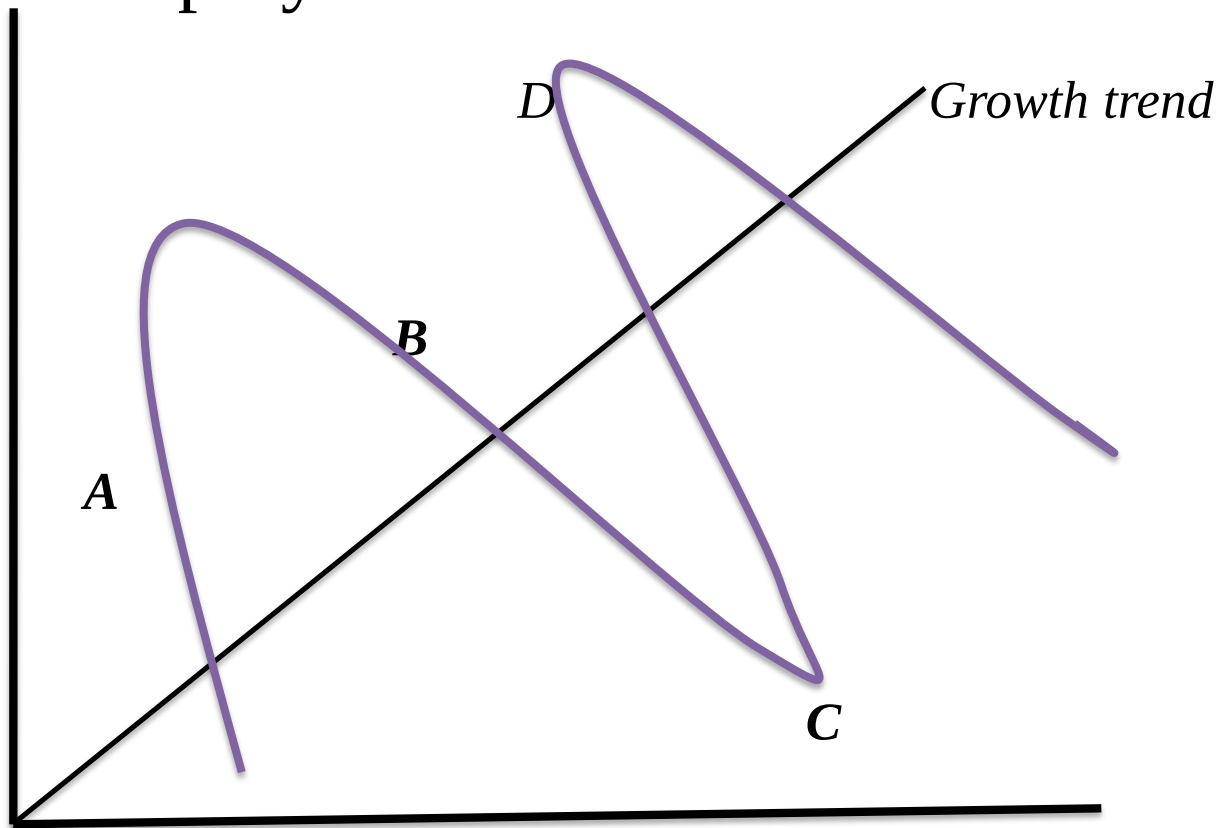
## ***3. Trough or Depression***

- ➡ Lowest point in business cycle
- ➡ The economy reached at low point
- ➡ unemployment rate is too high but total output reaches to low
- ➡ It is the severe stage of recession.

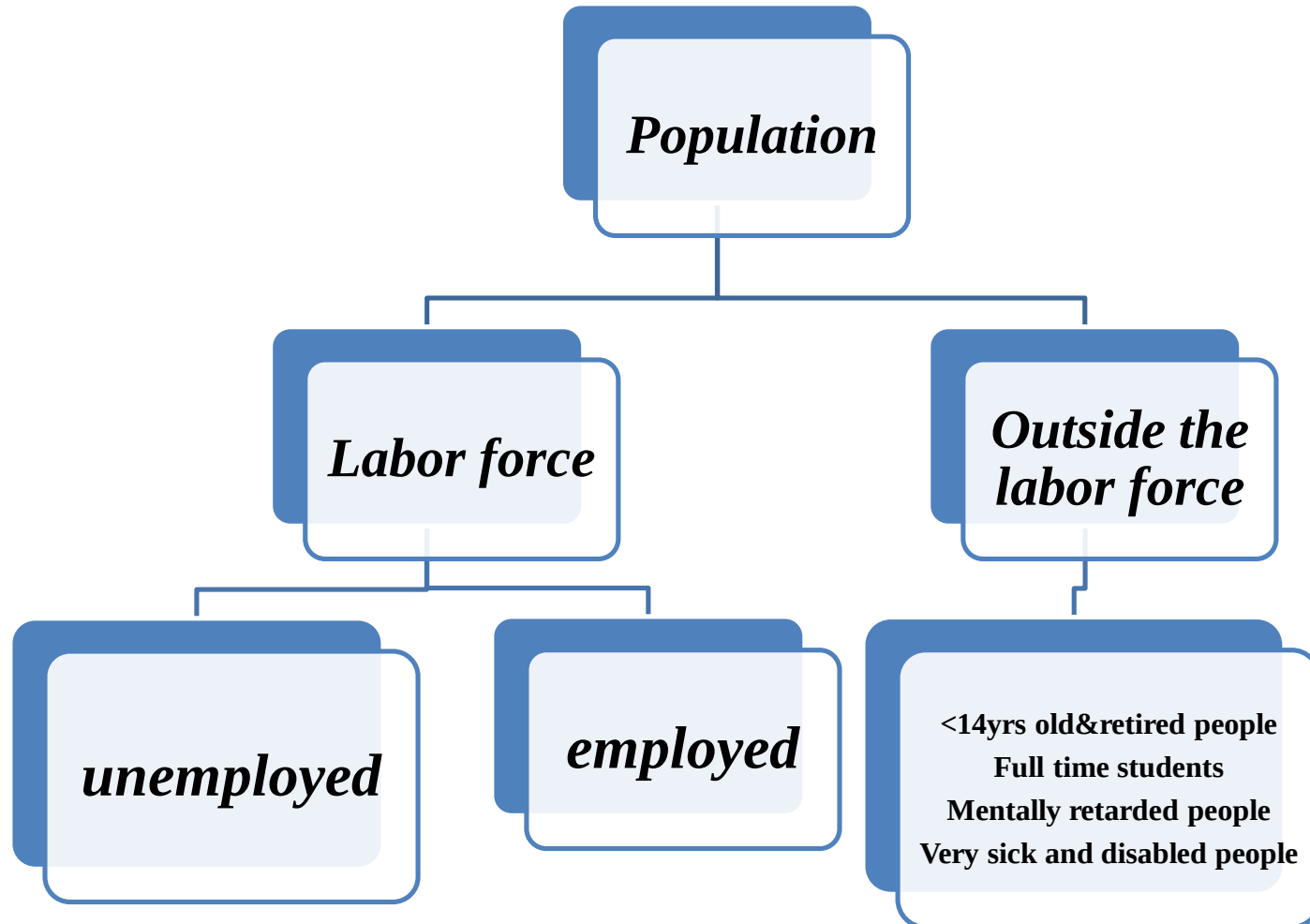
# Cont...

## 4. *Recovery*

- ➡ the economy starts to grow or recover
- ➡ total output starts to raise, national income rises and unemployment starts to decline



## 2. Unemployment



# Cont...

***Labor force = Employed + Unemployed***

***Unemployment*** is a portion of the labor force that is without a job but is actively searching for a job.

## Types of Unemployment

Commonly ,there are three types of unemployment

➡ ***Frictional unemployment***

➡ ***Structural unemployment***

➡ ***Cyclical unemployment***

# Cont...

## 1. *Frictional unemployment*

- ✚ refers to brief period of unemployment experienced by people ***due to the following reasons:***
- ➡ Seasonality of work.eg. Due to bad weather,...
- ➡ Entrance to the labor force. If a student spends few months looking for a jobs after graduation, he is considered as frictionally unemployed.
- ➡ Voluntary switching of job, either in search of better jobs or the previous job doesn't work out.
- ➡ Re-entering to the labor force

## 2. Structural Unemployment

- ✚ refers to unemployment resulting from mismatch between skills and locations of job seekers and location of vacancies
- ✚ Eg. A nuclear physicist may not get job in Ethiopia
- ✚ Typist may not get job in the environment where computers are not used for work

*Structural Unemployment can be caused by*

- Permanent shifts of demand for goods and services
- Technological change- automations and computerization of work may force the unskilled workers who used to produce the product manually to be unemployed

# 3.Cyclical Unemployment

- ✚ Unemployment generated due to the absence of jobs and vacancies
- ✚ Happens due to deficiency in demand for goods and services when the economy is in ***recession***
- ✚ Frictional and structural unemployment are known as ***natural level of unemployment, because they are unavoidable.***



# Measuring Unemployment

✚ Measured by *unemployment rate*

✚  $TU = FU + SU + CU$

$$UR = TU/LF$$

$$NR = (FU + SU)/LF, \text{ where:}$$

TU-total employment                      SU-structural unemployment

CU-cyclical unemployment    FU- frictional unemployment

NR-natural rate of unemployment    LF-labor force

❖ Full employment doesn't mean zero employment

❖ *Economic cost of unemployment* is the amount of output that could have been produced had the unemployed people been employed. It is the amount of the forgone output.

# 3. Inflation

- ➡ Is *sustainable or continuous* increase in the general(*average*) price level of goods and services
- ➡ can be ***anticipated***(*expected to occur*) or ***unanticipated***(*actual inflation – expected inflation*)
- ➡ can be measured using price index or GDP deflator
- ➡ Price index refers to consumer and producer index

$$\text{PI} = \frac{\text{current cost of a mkt basket of g+s}}{\text{cost of the same mkt basket in the base yr}} \times 100$$

$$\text{CPI this yr} = \frac{\text{current cost of a mkt basket of consumer g+s}}{\text{cost of the same mkt basket at the base yr}} \times 100$$

- The base yr is a reference year and it is usually a normal year when there is neither natural nor man made catastrophe.

## Cont...

- Inflation rate at time  $t = [(CPI_t - CPI_{t-1}) / (CPI_{t-1})] * 100$

Example: consider that the consumer price index in Ethiopia in 1992 E.C was 120 while in 1991 E.C it was 100.  
calculate the inflation rate in 1992?

$$\begin{aligned}\text{Inflation rate in 1992} &= [(CPI_{1992} - CPI_{1991}) / (CPI_{1991})] * 100 \\ &= [(120 - 100) / 100] * 100 \\ &= 20\%\end{aligned}$$

# Causes of inflation

***Generally, there are two causes of inflation***

➡ Demand – pull inflation

➡ Cost-push inflation

***Demand – pull inflation***

➡ resulting from an increase in demand for goods and services while the economy is producing at or close to full employment

➡ It happens when the economy attempts to spend beyond its production capacity

➡ Aggregate DD grows greater faster than economy's productive capacity at full employment (aggregate SS), which creates shortage and this pushes price upward.

# Cont...

## ***Cost -push inflation***

- ➡ Arises due to a continuous decline in aggregate supply of goods and services. This may happen due bad weather condition, an increase in wage and price of other inputs, decline in productivity

## **Effects of inflation**

There are two important effects of inflation:

### ***Redistributive effect and output effect***

#### ***1. Redistribution effects of inflation***

- ➡ ***Price effect*** – those people who consume those g+s whose price are rising rapidly suffer more than those who consume g+s whose price rise slowly.
- ➡ ***Income effect*** - increase in price reduce the purchasing power
  - \* *redistribute income from fixed income earners to flexible income earners*
  - \* *harm savers and creditors and benefit borrowers*
- ➡ ***wealth effects***- people who own assets like land and BLDG will be better off than others b/c the real value of such asset doesn't decrease during inflationary periods.

# Cont...

## ***2. Output effects of inflation***

- ➡ it creates uncertainty – difficult for producer, investors to make production decisions
- ➡ Encourage speculation
- ➡ Shorten time horizon- during inflationary period producers, sellers and consumers postpone or cancel their expenditure plans. This may reduce the total output and employment level in the economy

# ***CHAPTER SEVEN***

## ***NATIONAL INCOME ACCOUNTING***

# National income account

- Is an accounting record of the level of economic activities of an economy
- is a measure of aggregate output, income and expenditure in an economy

## ***Its importance:***

- To know economic performance of country
- To observe the long run trend of the economy
- To formulate economic policies



# Gross Domestic Product(GDP) and Gross National Product(GNP)

## ***GDP***

➡ is the ***market of currently produced final*** goods and services that are produced ***within in the country's borders(boundary)*** during ***a given period of time***, usually one year.

## ***GNP***

➡ is the ***market value of currently produced final*** goods and services that are produced by domestically owned factors of production during ***a given period of time***

➡ It measure the total output or income generated by domestic residents' factors of production irrespective of where these resource are used.

# Cont...

$$\mathbf{GNP = GDP + NFI}$$

NFI- net factor income

NFI = factor income received from abroad - factor income received by foreigners from domestic economy

NFI can be positive, negative or zero

$NFI > 0$ , then  $GNP > GDP$

$NFI < 0$ , then  $GNP < GDP$

$NFI = 0$ , then  $GNP = GDP$

# Problems associated with measuring GDP

## *The problems:*

✓ Double counting

To avoid this problem we should take

- The market value of final g+s
- The value added at each stage of production
- ✓ Don't include non-productive transactions...transfer payments
- ✓ Failing to measure non-marketed outputs since they have no market price
- government production of public goods
- households output which produced for home consumption purpose
- Leisure activities such as the satisfaction obtained from recreational activities aren't included in GDP

# Cont.....

- ✓ It is difficult to measure outputs in the informal sector
- the informal sector is that part of the economy which consists of activities or output that are never report to tax and other gov't authorities. It include both legal(sealing tela, tej,...) and illegal(like drug dealing) activities
- ✓ Distribution of income across household is not included in GDP
- ✓ Cost of environmental damage is not counted in GDP.

# Measuring GDP and GNP

Three methods:

1. *Product approach/value added approach*
2. *Expenditure approach*
3. *Income approach*

## **1. Product approach**

- ✓ *Calculated by adding the market value of goods and services produced by each sector of the economy i.e. agriculture, service and industry*

# Cont...

## 2.Expenditure Approach

- + **GDP** is measured by adding all spending on final final goods and services produced in the country
- + The expenditures made on the final goods and services of country can be divided into ***four categories***.

These are

- I. Personal consumption expenditure
- II. Gross domestic private investment expenditure
- III. Government spending on goods and services
- IV. Net export

# Cont...

## ***I. Personal consumption expenditure(C)***

- Expenditure made by households to purchase final g+s for consumption purpose
- They are divided into three:
  1. Durable consumer goods- are those goods that can be used for longer period of time such as cars, TV,...
  2. Non-durable consumer goods- are those goods that can be used/consumed for shorter period of time such as food, fuel,...
  3. Services- include education, haircut, transport, health service....

# Cont...

## ***II. Gross domestic private investment expenditure(I)***

➤ Expenditure by private sectors for investment purpose  
 $GI = \text{Net Inv't} + \text{depreciation}(\text{consumption of fixed asset})$

➤ Includes three types of expenditures

1. Business investment expenditure/business fixed investment on machinery, equipment

2. Construction expenditure-expenditure on residential and non residential constructions

3. All increase in inventories-these is unsold output currently produced by firms and counted in GDP as if they were purchased by the firms' themselves



# Cont...

## III. *Government spending on goods and services(G)*

- Expenditure by local and federal government
- NB:All expenditures made by government will not be counted.

Eg. Public transfer payments like pension and interest payments on the country's debt will not be counted

# Cont...

## ***IV. Net export(NX)***

- It means export ( $X$ ) minus import ( $M$ )
- Imported goods are not our country's output, these we can not include them in GDP calculation

$$\mathbf{GDP = C+I+G+NX}$$

# Cont...

## ***3. Income Approach***

- GDP is measured by adding all incomes earned by domestic factors of production to produce goods and services
- These factors of production are labor, land , capital, and entrepreneurship

# Cont...

***These payments includes:\***

- i. Compensation of employees: the reward for labor. i.e. Wages and salaries
- ii. Rental income: the reward for land
- iii. Proprietor's income/ profits/: income earned by entrepreneurs.
- iv. Interest income: the reward for capital
- v. Depreciation: allowance of capital consumption  
A portion of capital goods used in the production process that wear out or used up in the current production process
- vi. Indirect business taxes: like excise and sales tax

$$\text{GDP} = \text{i} + \text{ii} + \text{iii} + \text{iv} + \text{v} + \text{vi}$$

# Nominal and Real GDP

## ***Nominal GDP***

- ➡ Is monetary value of currently produced final outputs measured at current market price
- ➡ Is GDP measured at current market price
- ➡ Is not useful to compare country's economic performance through time

## ***Real GDP***

- ➡ it is the value of currently produced g+s at constant price. or it is measure of GDP at constant price
- ➡ It is calculated by dividing the nominal GDP by price index

$$\text{Real GDP} = (\text{nominal GDP} / \text{price index}) \times 100$$

# Cont...

Eg. The Ethiopian nominal GDP and the CPI for the year 1985 and 1986 was given below. Calculate the real GDP and the economic growth?

| Year | nominal GDP | CPI   | real GDP |
|------|-------------|-------|----------|
| 1985 | 26,671.40   | 849.9 | 3138.18  |
| 1986 | 28,328.90   | 858.9 | 3298.28  |

$$\text{Economic growth} = \frac{\text{RGDP}_t - \text{RGDP}_{t-1}}{\text{RGDP}_{t-1}} \times 100$$
$$5.1 \%$$

# Other social account

- **Net domestic product (NDP):** it is the measure of the total value of the economy without impairing the future production capacity of the economy.

$$\text{NDP} = \text{GDP} - \text{Depreciation}$$

- **National Income (NI):** is the total income earned by our factors of production irrespective of where they are used.

$$\text{NI} = \text{NDP} + \text{NFI} - \text{indirect business tax}$$

- **Personal income:** income earned by persons or hhs.

$$\text{PI} = \text{NI} - [\text{social security contribution} + \text{corporate income tax} + \text{retained corporate profit}] + [\text{public transfer payments} + \text{net interest on government debt}]$$

- **Personal disposable income(Yd):** home take money

$$\text{Yd} = \text{PI} - \text{personal taxes}$$

$$\text{Yd} = c + s$$

# ***Income inequality and Distribution***

GDP and GNP doesn't indicate how the total output currently produced is distributed in the economy

The distribution of income measured in two measures

## ***i. Functional distribution of income***

- ✓ refers to the distribution of national income to owners of factors of production, i.e. labor, land, ...

## ***ii. Personal distribution of income***

- ✓ Shows the distribution of national income to individual families



# Cont...

***The sources of income inequalities in country can be :***

- i. Differences Wealth
- ii. Differences Ability
- iii. Discrimination: which leads to difference in opportunities for achievements.
- iv. Differences in education, energy and motivation
- v. Differences in human capital and investment choices

# GDP and Informal sector

- ***Informal sector*** of the economy is part of the economy which consists of activities that are never reported to government authorities
- These activities can be ***legal*** or ***illegal***
- GDP doesn't usually include these activities due to lack of information
- Tax evasion is usually thought to be the major incentive for a person to participate in the informal sector.
- When the size of informal sector is large , the GDP understates the performance of the economy

# Exercise

1. Suppose that the year 2004 exports were equal to imports and gross investment was equal to government expenditure. Private final consumption expenditure was twice its government expenditure and equal to birr 300 million.
  - a. Calculate GDP
  - b. Calculate GNP assuming NFI was birr 30 million
  - c. Calculate the value of depreciation if the value of NNP during the year was birr 430 million?

2. Given

$$C = 200 + 0.02Y_d$$

$$G = 500, \quad I = 400, \quad NX = 200 \quad \text{and} \quad Y_d = 0.25GDP$$

Find GDP,  $Y_d$ , C